

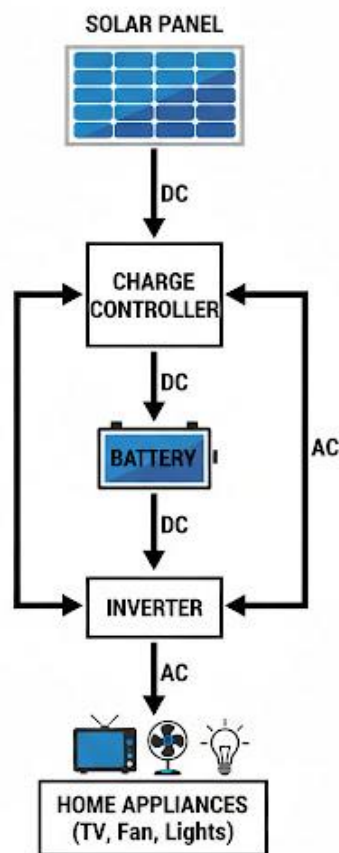
FIRST TERM - JSS 2 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE LESSON NOTES

WEEK 1: REVIEW OF JSS 1 WORK

Topic: Review of JSS 1 Work

Breakdown (Subtopics):

- Revision of basic concepts in solar PV systems
- Review of components and their functions
- Safety precautions during installation
- Importance of entrepreneurship in solar PV



Performance Objectives: By the end of this lesson, students should be able to:

1. Define Solar Photovoltaics (PV) and its purpose.
2. Recall and list the four primary components of a standalone solar PV system.
3. State the specific function of each of the four components.
4. List at least five crucial safety precautions for solar installation.
5. Explain two reasons why entrepreneurship in solar PV is important in Nigeria.

Lesson Content:

1. Basic Concepts: What is Solar PV?

- **Definition:** "Solar" refers to the sun. "Photo" comes from the Greek word *phos*, meaning light. "Voltaic" refers to electricity, named after Alessandro Volta, an pioneer of electricity.
- Therefore, a **Solar Photovoltaic (PV) system** is a technology that *directly converts sunlight (photons) into electricity (voltage)*.
- This is *different* from Solar Thermal, which uses the sun's *heat* to warm water. Solar PV creates usable electricity.

2. Review of Primary Components and Functions A basic standalone (off-grid) solar system has four essential parts:

1. Solar Panels (The "Collector"):

- **Function:** This is the "engine" of the system. The panels are made of many photovoltaic cells (usually silicon). When sunlight strikes the panel, it excites electrons, causing them to flow. This flow is **Direct Current (DC)** electricity.

2. Charge Controller (The "Regulator/Brain"):

- **Function:** This is the "brain" or "traffic warden" of the system. It is placed between the solar panels and the batteries. Its job is to manage the flow of electricity.
- **Two Main Jobs:**

- **Prevents Overcharging:** It stops the panels from sending power to the batteries when the batteries are full, preventing damage (like boiling or explosion).
- **Prevents Over-discharging:** It stops the battery from supplying power to appliances when the battery is almost empty, which would permanently kill the battery.

3. **Battery (The "Bank/Storage"):**

- **Function:** To store the DC electricity produced by the panels during the day so it can be used at night or on cloudy days. Solar panels only work when the sun is shining, but we need electricity 24/7. The battery bridges this gap.

4. **Inverter (The "Converter"):**

- **Function:** This device *inverts* (changes) the type of electricity. The panels and batteries use **Direct Current (DC)**, but most home appliances (TVs, fans, bulbs, fridges) use **Alternating Current (AC)**.
- The inverter's job is to convert the low-voltage DC power from the battery into high-voltage AC power that your appliances can use.

3. Safety Precautions (Recap) Working with electricity is dangerous, even if it comes from the sun.

1. **Never Work in the Rain:** Water and electricity are a deadly combination.
2. **Wear Personal Protective Equipment (PPE):** Always wear safety goggles (to protect eyes), rubber-soled boots, and insulated gloves.
3. **Avoid Electric Shock:** Solar panels produce live DC electricity even in low light. Never touch broken panels or bare wires.
4. **Battery Safety:** Batteries contain acid and can release explosive gas. Never smoke near batteries, and always connect them in a well-ventilated area.
5. **Short Circuits:** Never let the positive (+) and negative (-) wires from a panel or battery touch each other. This will create a very hot, dangerous spark (an arc flash) that can cause fires or blindness.
6. **Work with a Partner:** Never work alone on a "live" system.

4. Importance of Entrepreneurship in Solar PV An **entrepreneur** is someone who starts a business to solve a problem.

- **The Problem:** Nigeria has an unreliable public electricity grid (NEPA/PHCN). Many rural areas have no grid electricity at all.
- **The Solution:** Solar PV provides a clean, reliable, and increasingly cheap alternative source of power.
- **The Business Opportunity (Importance):**
 1. **High Demand:** Millions of homes and businesses in Nigeria need reliable power. This creates a massive market for solar installers, sellers, and maintenance technicians.
 2. **Job Creation:** The solar industry creates "green jobs" for people (like you) who learn this skill, helping to reduce unemployment.
 3. **Self-Reliance:** Learning solar installation is a high-value vocational skill that allows you to be self-employed (be your own boss) and create your own company.
 4. **Environmental Impact:** As a solar entrepreneur, you are actively helping to solve Nigeria's power problem *and* protect the environment by reducing the noise and pollution from petrol generators.

Evaluation:

1. What does "Photovoltaic" mean?
2. What are the four main components of an off-grid solar system?
3. What is the function of an inverter?
4. What are the two main jobs of a charge controller?
5. State three safety rules you must obey during installation.

Assignment:

1. In your own words, what is a solar PV system?
2. Draw a simple block diagram showing the four main components and how they are connected.
3. Why is it dangerous to let the positive and negative wires from a battery touch?

4. Explain two reasons why being a solar installer is a good business idea in Nigeria.
5. What is the difference between Solar PV and Solar Thermal?

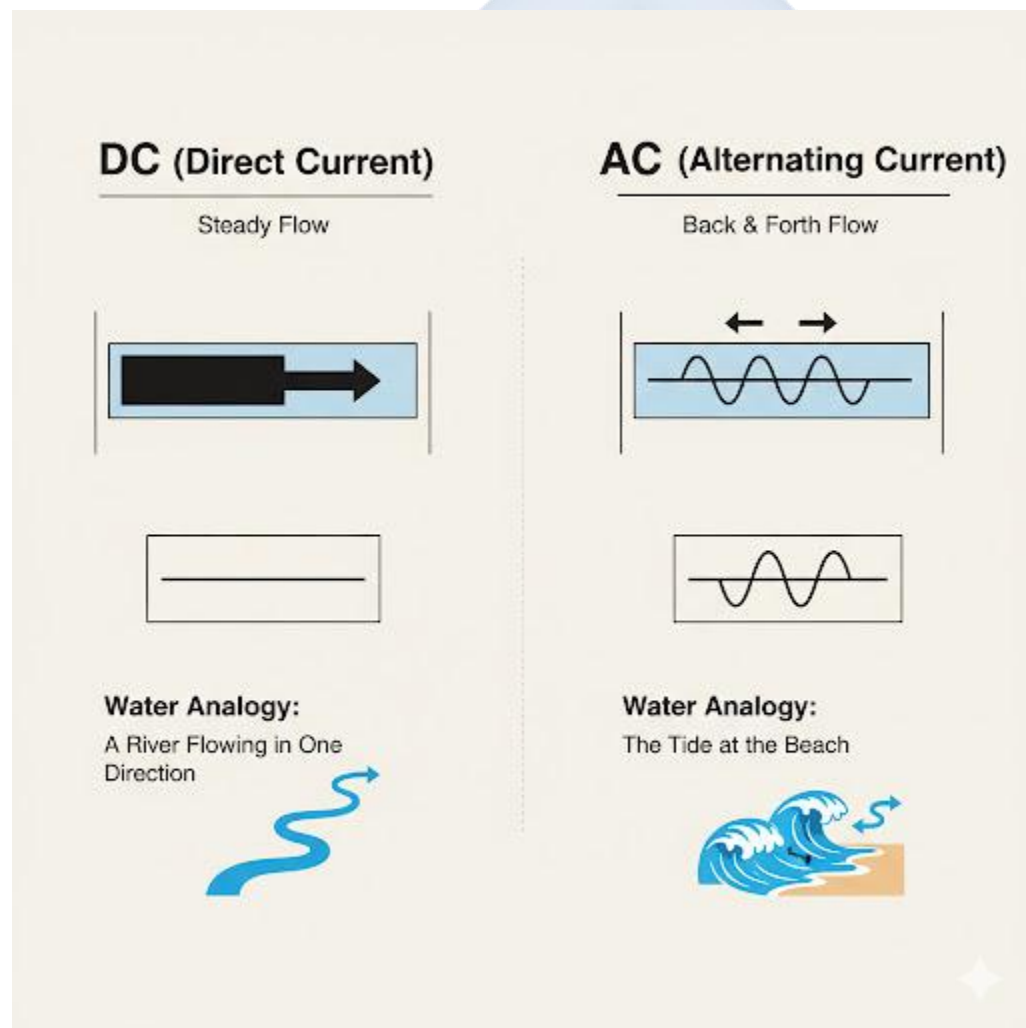


WEEK 2: ELECTRICITY FUNDAMENTALS

Topic: Electricity Fundamentals

Breakdown (Subtopics):

- Definition and importance of electricity
- Conductors and insulators
- Types of current: AC and DC
- Voltage, current, and resistance (Ohm's Law)
- Measuring electrical quantities using a multimeter



Performance Objectives: By the end of this lesson, students should be able to:

1. Define electricity.
2. Differentiate between a conductor and an insulator with two examples of each.
3. Differentiate between Alternating Current (AC) and Direct Current (DC) and state the source of each.
4. Define Voltage, Current, and Resistance using the "water pipe" analogy.
5. State Ohm's Law ($V=IR$).
6. Identify a multimeter and its basic functions.

Lesson Content:

1. Definition and Importance of Electricity

- **Definition:** Electricity is a form of energy that results from the flow of tiny, negatively charged particles called **electrons**.
- **Importance:** Modern society runs on electricity. It powers our homes (lights, TV, fans, fridges), hospitals (life-support machines), schools (computers), factories, and communication (phones, internet).

2. Conductors and Insulators

- **Conductors:** These are materials that **allow electricity (electrons) to flow** through them easily. They have low resistance.
 - *Examples:* Metals (Copper is the best, Aluminum, Gold, Silver), Water (especially salty water), the human body.
- **Insulators:** These are materials that **block or stop the flow** of electricity. They have high resistance.
 - *Examples:* Rubber, Plastic, Glass, Dry Wood, Ceramic.
- **Application:** We use both. We use a *conductor* (copper) for the inside of a wire and an *insulator* (plastic) for the outside to protect us from shock.

3. Types of Current: AC and DC (Crucial for Solar) This is the most important concept in solar installation.

- **A. DC (Direct Current):**

- **Definition:** Electricity that flows in **only one direction** (Direct), from negative to positive.
- **Analogy:** A river flowing steadily in one direction.
- **Sources:** **Solar Panels, Batteries**, and your phone/laptop charger (which converts AC to DC).

- **B. AC (Alternating Current):**

- **Definition:** Electricity that rapidly **alternates its direction**, flowing back and forth, 50-60 times per second (50Hz in Nigeria).
- **Analogy:** The tide at the beach, pushing in and pulling out.
- **Sources:** The public grid (**NEPA/PHCN**), petrol **Generators**, and **Inverters** (which *create* AC from DC).

- **Why it matters:** You *must* connect DC to DC (panel to battery) and AC to AC (inverter to TV). Connecting them incorrectly will destroy the appliance.

4. Voltage (V), Current (I), and Resistance (R) We use the **water pipe analogy** to understand these three concepts.

- **Voltage (V):**

- **Definition:** This is the **electrical pressure** or "push" that makes the electrons flow.
- **Analogy:** The *pressure* in the water pipe.
- **Unit: Volts (V).** (e.g., A car battery is 12V).

- **Current (I):**

- **Definition:** This is the *volume* or *flow rate* of electrons passing a point per second.
- **Analogy:** The *amount* of water flowing through the pipe (how fast it's flowing).
- **Unit: Amperes (Amps) (A).** (e.g., A 5A fuse).

- **Resistance (R):**

- **Definition:** This is the **opposition** to the flow of current. It "resists" the flow, creating heat.
- **Analogy:** A *narrowing* or *blockage* (like sand) in the water pipe.
- **Unit: Ohms (Ω).** (e.g., A 100Ω resistor).

5. Ohm's Law

- **Definition:** Ohm's Law describes the mathematical relationship between V, I, and R.
- **The Formula:** $V = I \times R$ (Voltage = Current multiplied by Resistance)
- **Simple Explanation:** "To push a high *volume of water* (I) through a *narrow pipe* (R), you need very high *pressure* (V)." This is why thin wires (high R) get hot when you pass a large current (I) through them.

6. The Multimeter

- **Definition:** A multimeter is the *most important* tool for a solar installer. It is a diagnostic tool used to **measure** electrical quantities.
- **Functions:** It can be set to measure:
 - **Voltmeter (V):** Measures Voltage (e.g., "Is this 12V battery charged?").
 - **Ammeter (A):** Measures Current (e.g., "How many amps is this panel producing?").
 - **Ohmmeter (Ω):** Measures Resistance (e.g., "Is this wire broken?").

Evaluation:

1. What is electricity?
2. What is the difference between a conductor and an insulator? Give one example of each.
3. What is the difference between AC and DC?
4. What is the source of AC power? What is a source of DC power?
5. Using the water pipe analogy, define Voltage, Current, and Resistance.

Assignment:

1. What does a multimeter do? List the three things it measures.

2. What is Ohm's Law (state the formula)?
3. Which type of current (AC or DC) is produced by a solar panel?
4. Which type of current (AC or DC) does your TV use?
5. What device is used to change DC to AC?

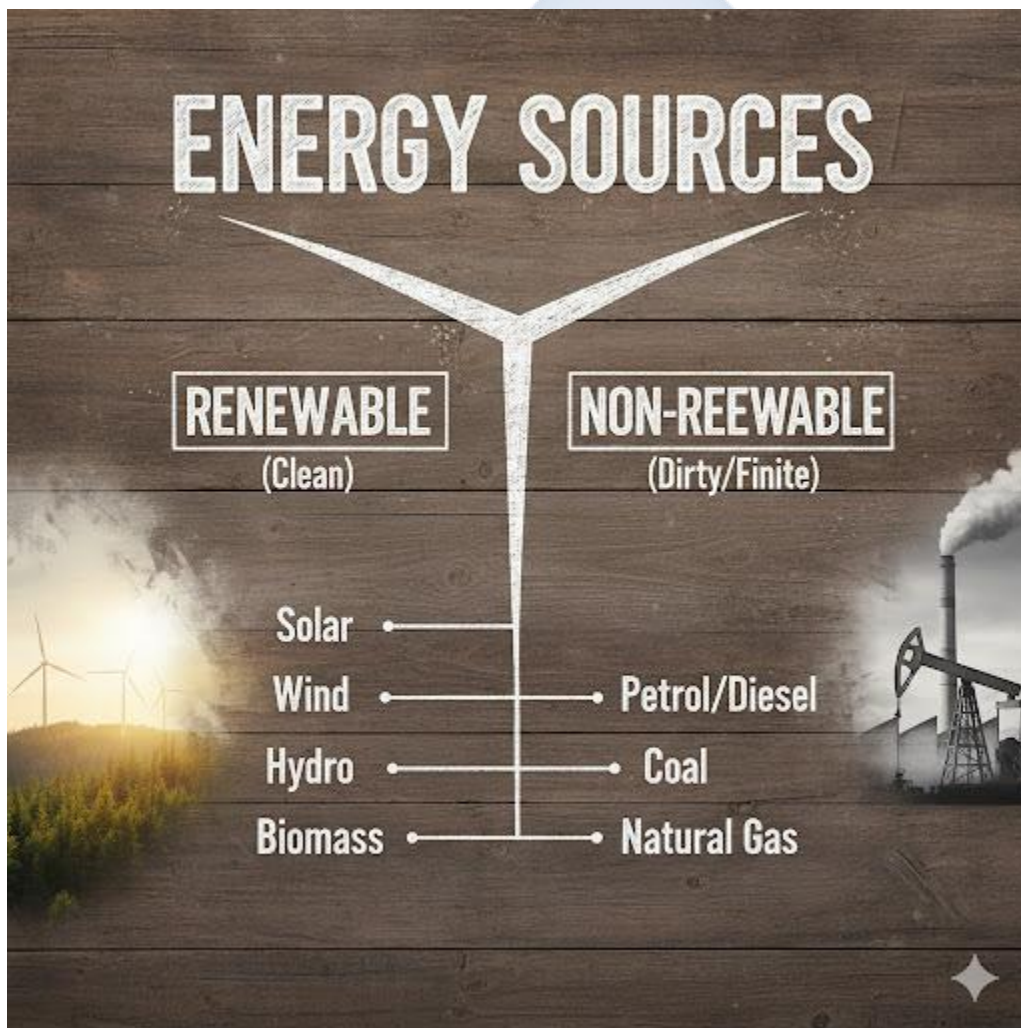


WEEK 3: ENERGY SOURCES AND POWER GENERATION

Topic: Energy Sources and Power Generation

Breakdown (Subtopics):

- Renewable vs Non-renewable energy
- Solar, wind, hydro, and thermal power
- Electricity generation process
- Importance of clean energy to the environment



Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between renewable and non-renewable energy sources.
2. List three examples of each type of energy source.
3. Briefly explain how solar, wind, and hydro power generate electricity.
4. Explain the importance of clean energy (renewable) for the environment.

Lesson Content:

1. Renewable vs. Non-Renewable Energy All energy on Earth comes from different sources.

We divide them into two groups:

- **A. Non-Renewable Energy (Dirty/Finite):**
 - **Definition:** These are energy sources that **cannot be replaced** in a short amount of time. Once we use them, they are gone forever (or take millions of years to form). They are *finite*.
 - **Problems:** They cause pollution and climate change.
 - **Examples:**
 1. **Fossil Fuels** (Crude Oil, Petrol, Diesel, Kerosene)
 2. **Natural Gas**
 3. **Coal**
- **B. Renewable Energy (Clean/Infinite):**
 - **Definition:** These are energy sources that **do not run out**. They are naturally replenished (refilled) by the Earth. They are *infinite*.
 - **Benefits:** They are clean and do not harm the environment.
 - **Examples:**
 1. **Solar** (from the sun)
 2. **Wind** (from moving air)
 3. **Hydro** (from moving water, like rivers or dams)
 4. **Geothermal** (from heat inside the Earth)
 5. **Biomass** (from burning organic plant matter)

2. Electricity Generation Process (How they work)

- **A. Solar (Photovoltaic):**

- **Process:** (Review) Uses **PV cells**. Sunlight (photons) hits the panel, and the energy knocks electrons loose, creating a flow of DC electricity.
- **No moving parts.**
- **B. Wind, Hydro, and Thermal (Turbine-Based):**
 - These three all use the same basic principle: **Spinning a Turbine**.
 - A **turbine** is a large machine with blades, like a fan. When it spins, it turns a **generator**, which produces electricity. The only difference is *what* spins the turbine.
 - **Wind Power:** The **wind** pushes the blades and spins the turbine.
 - **Hydro Power: Falling water** (released from a dam, like Kainji Dam) is channelled to push and spin the turbine.
 - **Thermal Power (Non-renewable): Coal or gas** is *burned* to boil water. This creates high-pressure **steam**, and the steam is used to spin the turbine.

3. Importance of Clean Energy to the Environment

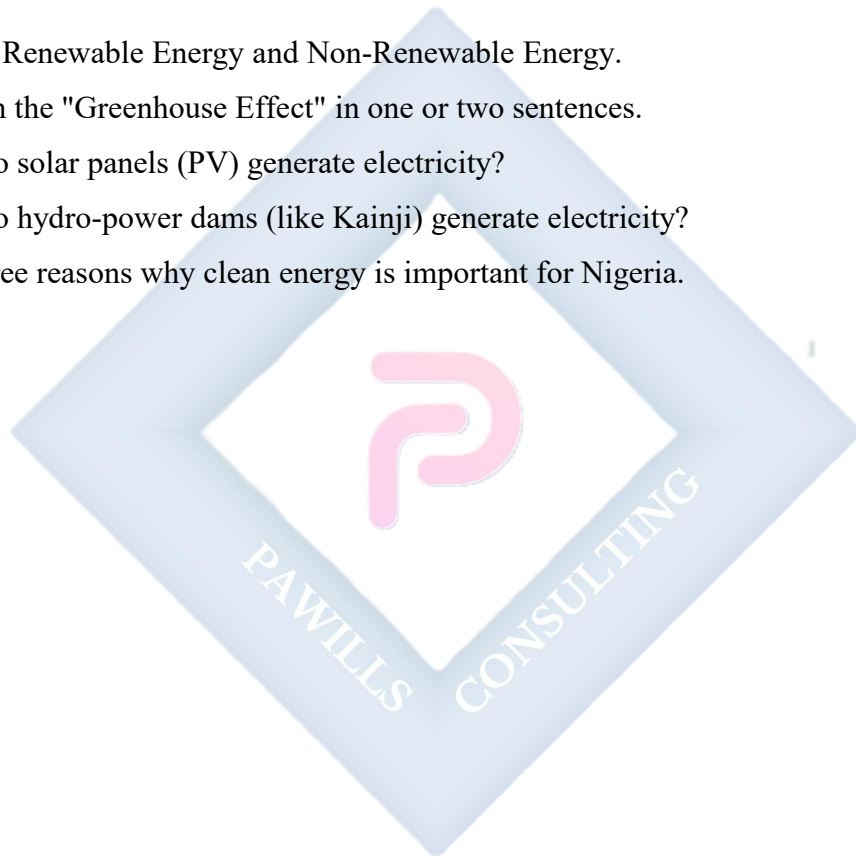
- **The Problem with Non-Renewable (Dirty) Energy:** When we burn petrol (in cars/generators) or gas (for power), it releases a gas called **Carbon Dioxide (CO₂)**.
- **The "Greenhouse Effect":** This CO₂ gas traps heat in the Earth's atmosphere, like a blanket. This is called the **Greenhouse Effect**, and it leads to **Global Warming** and **Climate Change**.
- **Importance of Clean Energy:**
 1. **Stops Climate Change:** Renewable sources (solar, wind, hydro) generate electricity *without* producing any CO₂. This helps stop global warming.
 2. **Reduces Air Pollution:** Burning fossil fuels also releases soot and chemicals that cause sickness (like asthma). Solar power is silent and clean, leading to better public health.
 3. **Sustainability:** Crude oil will one day run out. The sun will not. Clean energy is sustainable for future generations.

Evaluation:

1. What is the main difference between a renewable and a non-renewable energy source?
2. List two examples of non-renewable sources.
3. List three examples of renewable sources.
4. What gas, released by burning petrol, causes global warming?
5. What common machine is used in wind, hydro, and thermal power plants to generate electricity?

Assignment:

1. Define Renewable Energy and Non-Renewable Energy.
2. Explain the "Greenhouse Effect" in one or two sentences.
3. How do solar panels (PV) generate electricity?
4. How do hydro-power dams (like Kainji) generate electricity?
5. List three reasons why clean energy is important for Nigeria.



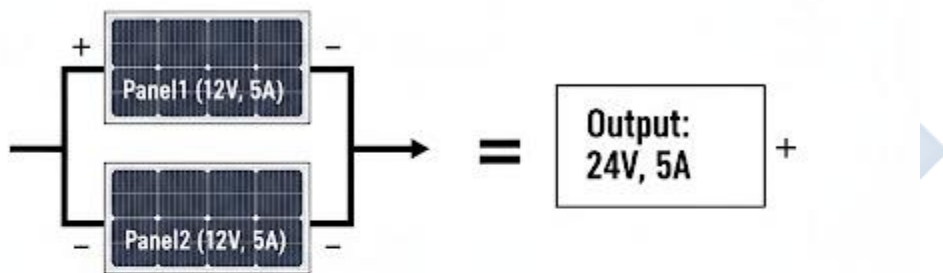
WEEK 4: SOLAR PV SYSTEM CONFIGURATION

Topic: Solar PV System Configuration

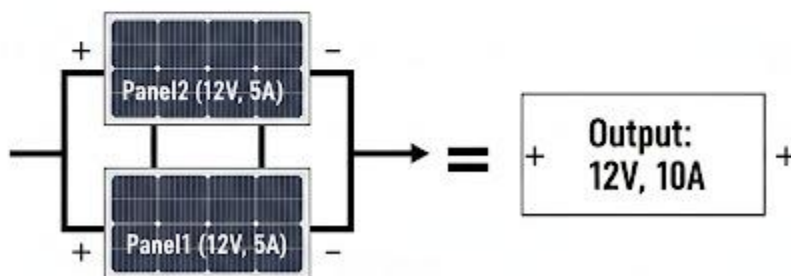
Breakdown (Subtopics):

- Types of solar PV systems: standalone, grid-tied, hybrid
- Components connection (series and parallel)
- Advantages and disadvantages of each configuration
- System setup demonstration

SERIES CONNECTION (Voltage Adds Up)



PARALLEL CONNECTION (Current Adds Up)



Performance Objectives: By the end of this lesson, students should be able to:

1. Define and differentiate between standalone, grid-tied, and hybrid solar systems.

2. Explain the key advantage of a hybrid system.
3. Define and diagram a "series" connection.
4. Define and diagram a "parallel" connection.
5. State the effect of a series connection on voltage and current.
6. State the effect of a parallel connection on voltage and current.

Lesson Content:

1. Types of Solar PV Systems There are three main ways to set up a solar system:

- **A. Standalone (Off-Grid) System:**
 - **Definition:** This system is **not connected** to the public electricity grid (NEPA/PHCN). It is independent.
 - **Components:** Solar Panels, Charge Controller, **Batteries** (essential), Inverter.
 - **Pros:** Provides power in remote areas with no grid. No monthly electricity bills.
 - **Cons:** You *must* buy expensive batteries. If your batteries are empty (e.g., after 3 cloudy days), you have no power at all.
- **B. Grid-Tied (On-Grid) System:**
 - **Definition:** This system is **connected** to the public grid. It does **not** have batteries.
 - **Components:** Solar Panels, Grid-Tied Inverter. (No controller, no batteries).
 - **How it works:** During the day, you use solar power. If you produce *more* power than you need, the excess is "sold" back to the grid (your meter spins backwards!). At night, you buy power from the grid.
 - **Pros:** Cheaper (no batteries), very efficient.
 - **Cons:** **If the grid (NEPA) goes down, your system also shuts down** (for safety). This makes it unsuitable for Nigeria.
- **C. Hybrid System:**
 - **Definition:** This is the **best of both worlds**. It is *connected* to the grid, but it *also* has batteries.
 - **Components:** Solar Panels, Charge Controller (or Hybrid Inverter), **Batteries**, Hybrid Inverter.

- **How it works:** It uses solar power first. If solar is not enough, it uses the battery. If the battery is empty, it uses the grid (NEPA) as a final backup.
- **Pros:** You have power 24/7. You save money by using your own power first.
- **Cons:** Most complex and expensive to set up.
- **This is the most common system for homes in Nigeria.**

2. Component Connection: Series vs. Parallel This is how you wire panels or batteries together to get the correct voltage and current.

- **A. Series Connection (Stacking Power):**

- **How:** Connect **Positive (+) to Negative (-)**, like stacking batteries in a torchlight.
- **Effect: VOLTAGE ADDS UP.** Current (Amps) stays the same.
- **Example:**
 - Panel 1 (12V, 5A) + Panel 2 (12V, 5A) in Series
 - **Result:** 24 Volts, 5 Amps.
- **Use:** Used to increase the voltage from the panels to match a 24V or 48V battery system.

- **B. Parallel Connection (Sharing Power):**

- **How:** Connect **Positive (+) to Positive (+)** and **Negative (-) to Negative (-)**. Like jump-starting a car.
- **Effect: CURRENT (Amps) ADDS UP.** Voltage stays the same.
- **Example:**
 - Panel 1 (12V, 5A) + Panel 2 (12V, 5A) in Parallel
 - **Result:** 12 Volts, 10 Amps.
- **Use:** Used to increase the *power* (Amps) going to the charge controller, or to increase the *capacity* (Amp-Hours) of your battery bank.

3. Demonstration: (Teacher draws the wiring diagrams for both Series and Parallel connections on the board).

Evaluation:

1. What is a "standalone" system? Where is it most useful?

2. What is a "hybrid" system? Why is it the most popular in Nigeria?
3. What is the main problem with a "grid-tied" system in Nigeria?
4. In a **series** connection (Panel 1 + Panel 2), what adds up: Voltage or Current?
5. In a **parallel** connection (Panel 1 + Panel 2), what adds up: Voltage or Current?

Assignment:

1. Define Standalone, Grid-Tied, and Hybrid systems.
2. Draw a simple diagram of a **series** connection for two batteries.
3. Draw a simple diagram of a **parallel** connection for two batteries.
4. You have two 12V, 100A batteries.
 - (a) What is the total voltage and amperage if you connect them in **series**?
 - (b) What is the total voltage and amperage if you connect them in **parallel**?
5. Which system type (standalone, grid-tied, hybrid) do you think is best for a hospital? Why?

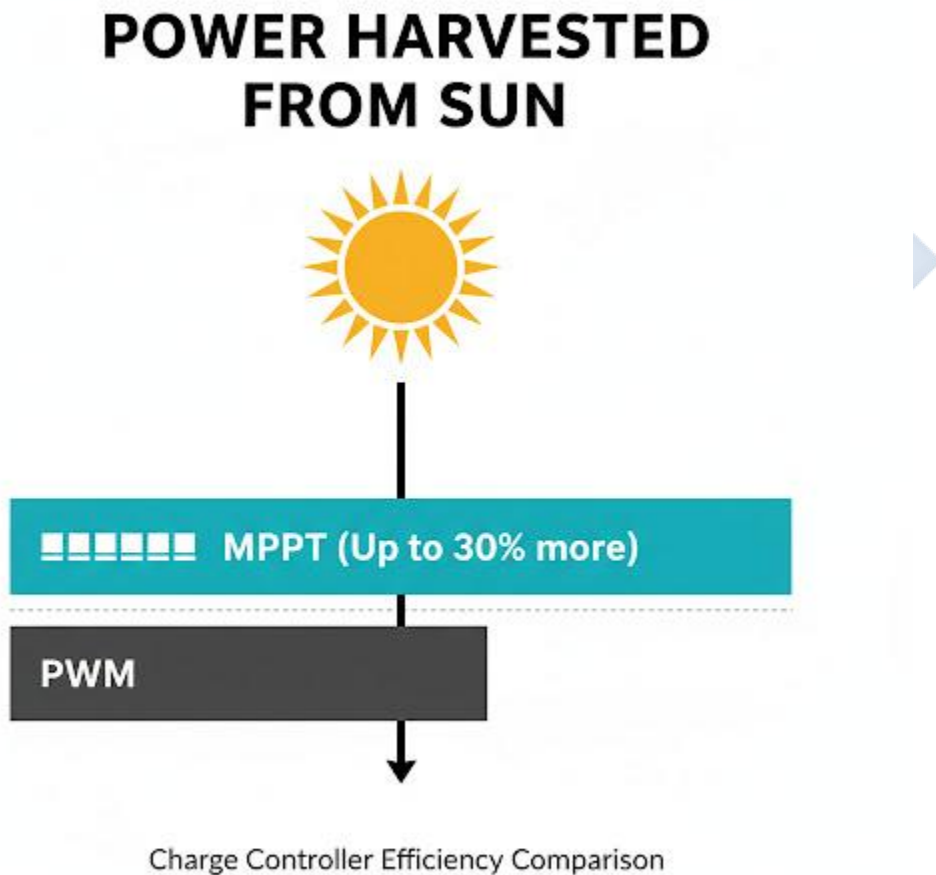


WEEK 6: SOLAR CHARGE CONTROLLERS

Topic: Solar Charge Controllers

Breakdown (Subtopics):

- Meaning and functions of charge controllers
- Types of charge controllers (PWM and MPPT)
- Importance of charge regulation
- Controller installation and wiring practice



Performance Objectives: By the end of this lesson, students should be able to:

1. State the primary role of a charge controller.

2. List the two main protective functions of a charge controller.
3. Differentiate between PWM and MPPT charge controllers.
4. State the most important rule for wiring a charge controller.

Lesson Content:

1. Meaning and Functions (Recap)

- **Definition:** The Solar Charge Controller is the "brain" or "regulator" of the system. It is a small box that is connected between the Solar Panels, the Batteries, and (sometimes) the DC Loads.
- **Main Function:** Its main job is to **protect the battery**. A battery is the most expensive and most easily damaged part of a solar system. The charge controller is its bodyguard.

2. Importance of Charge Regulation (The 2 Main Jobs):

1. Overcharge Protection:

- **Problem:** On a sunny day, the solar panels can keep forcing electricity into the battery even after it is 100% full.
- **Damage:** This "overcharging" will boil the acid (in lead-acid batteries) or damage the cells (in lithium batteries), destroying the battery.
- **Solution:** The controller *senses* the battery voltage. When it's full, the controller *stops* or *reduces* the power coming from the panels.

2. Over-discharge Protection (Low Voltage Disconnect):

- **Problem:** The inverter will keep pulling power from the battery to run appliances, even when the battery is almost empty.
- **Damage:** Draining a battery to 0% (deep discharging) will permanently damage it.
- **Solution:** The controller *senses* when the battery voltage is dangerously low and *cuts off* the power to the inverter/load, protecting the battery from dying.

3. Types of Charge Controllers

- **A. PWM (Pulse Width Modulation):**

- **Technology:** This is the *older, simpler, and cheaper* technology.
- **How it works:** It acts like a simple electronic switch. It connects the panel directly to the battery. When the battery gets full, it starts "pulsing" (switching on/off very fast) to reduce the power.
- **Disadvantage:** It is not very efficient. It forces the solar panel (which might be 18V) to operate at the battery's voltage (e.g., 12V), which *wastes* a lot of power.
- **B. MPPT (Maximum Power Point Tracking):**
 - **Technology:** This is the *newer, smarter, and more expensive* technology.
 - **How it works:** An MPPT controller is a "smart converter." It *tracks* the panel's "sweet spot" (the Maximum Power Point, or best voltage/current combination) and *converts* that power to the exact voltage the battery needs.
 - **Advantage:** It is highly efficient (up to 99%) and can get up to 30% *more* power from the same solar panels compared to a PWM controller. This is the modern standard for all serious systems.

4. Controller Installation and Wiring (Safety)

- **THE GOLDEN RULE: CONNECT THE BATTERY FIRST!**
- **Reason:** The charge controller needs to get its *power* and *voltage reading* from the battery *first*, so it knows if it is a 12V or 24V system. If you connect the panel first, the high voltage from the panel can damage the controller.
- **Steps:**
 1. Mount the controller on the wall (in a cool, dry place).
 2. Connect the controller's **BATTERY** terminals (+ and -) to the **BATTERY** terminals. (The controller's screen will light up).
 3. Connect the controller's **PV (or SOLAR)** terminals (+ and -) to the **SOLAR PANEL** terminals.
 4. Connect the controller's **LOAD** terminals (if used) to your DC lights/appliances.

Evaluation:

1. What is the main component that a charge controller is designed to protect?

2. What is "overcharge protection"?
3. What is "over-discharge protection"?
4. What is the main difference between a PWM and an MPPT controller?
5. What is the "Golden Rule" of wiring a controller? (Connect... first).

Assignment:

1. List the two main functions of a charge controller.
2. What do the acronyms PWM and MPPT stand for?
3. Why is an MPPT controller better than a PWM controller?
4. Why must you connect the battery to the controller *before* connecting the solar panel?
5. Draw a simple diagram showing a Panel, a Controller, and a Battery wired together correctly.

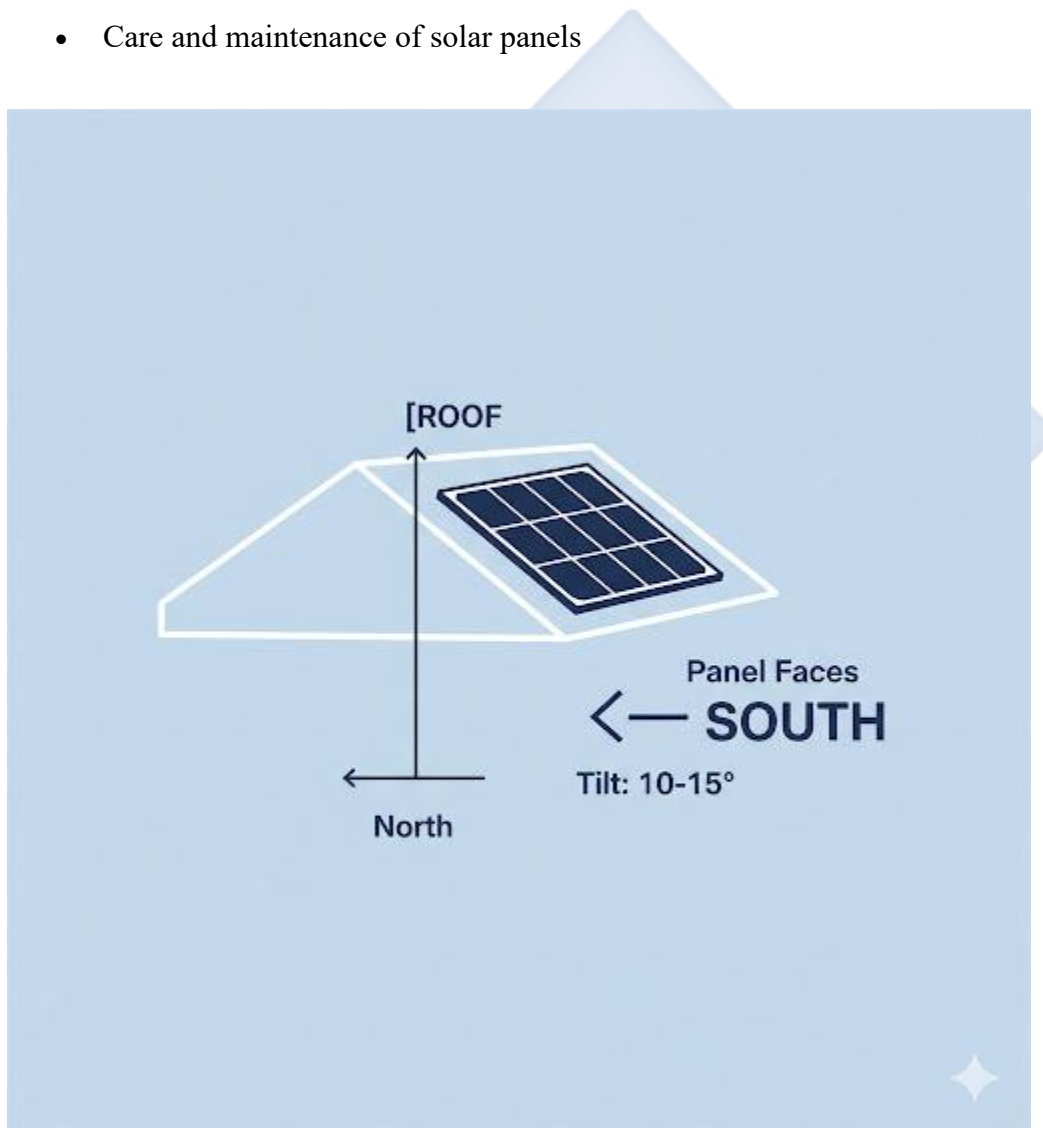


WEEK 8: SOLAR PANELS AND MOUNTING TECHNIQUES

Topic: Solar Panels and Mounting Techniques

Breakdown (Subtopics):

- Review of solar panels and their efficiency
- Steps in mounting solar panels
- Mounting angles and direction
- Care and maintenance of solar panels



Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between monocrystalline and polycrystalline panels.

2. State the correct *direction* (azimuth) a solar panel must face in Nigeria.
3. State the optimal *tilt angle* for a solar panel in Nigeria.
4. List the basic steps for mounting a panel on a roof.
5. Describe the correct way to clean and maintain solar panels.

Lesson Content:

1. Review of Solar Panels

- **Monocrystalline ("Mono"):**
 - **Look:** Black, with rounded-off corners on the cells.
 - **Efficiency:** *High* efficiency (18-22%). They produce more power in less space.
 - **Cost:** *High* cost.
- **Polycrystalline ("Poly"):**
 - **Look:** Blue, with a "marbled" or "broken glass" look. Cells are perfect squares.
 - **Efficiency:** *Standard* efficiency (15-18%).
 - **Cost:** *Cheaper*.

2. Panel Mounting Angles and Direction (CRITICAL) For a panel to produce its maximum power, it must get the most direct sunlight.

- **A. Direction (Azimuth):**
 - The Earth is a sphere, and Nigeria is in the **Northern Hemisphere**.
 - The sun spends all day in the *southern* part of our sky.
 - **Rule:** In Nigeria (and everywhere in the Northern Hemisphere), solar panels must **ALWAYS FACE TRUE SOUTH**. Facing them north, east, or west will *drastically* reduce power output.
- **B. Angle (Tilt):**
 - The tilt is the angle from the horizontal (flat ground).
 - The *perfect* tilt angle is equal to your location's latitude.
 - **Rule:** For Nigeria (which is between 4° and 14° latitude), a fixed tilt angle of **10 to 15 degrees** is optimal.
 - *Do not* mount panels flat (0°), as this allows water and dust to gather.

3. Steps in Mounting Solar Panels (Roof Mount)

1. **Inspect the Roof:** Ensure the roof is strong and free of damage.
2. **Install Mounts:** Securely bolt the "feet" or "L-brackets" of the mounting system to the roof rafters. This is the most critical step for safety.
3. **Seal the Roof:** Apply waterproof sealant (like tar or silicone) to every hole you drilled to prevent leaks.
4. **Assemble Racks:** Attach the aluminum rails to the mounting feet.
5. **Mount Panels:** Place the solar panels onto the rails.
6. **Secure Panels:** Use the "mid-clamps" (between panels) and "end-clamps" (at the ends) to bolt the panels tightly to the rails.
7. **Grounding:** Connect a copper "earth" wire from the metal frame of the panels to a grounding rod. This is a safety step.

4. Care and Maintenance of Solar Panels

- **1. Cleaning:** This is the most important maintenance.
 - **Problem:** Dust, bird droppings, and leaves can cover the panel and block the sun, reducing power by up to 30%.
 - **Solution:** Clean the panels every 4-6 weeks (or after a dusty/windy period).
 - **How:** Use a soft brush (with a long pole) and clean water. **NEVER** use harsh soap or an abrasive (scratchy) sponge. Clean in the *early morning* or *late evening* when the panels are cool (hot glass + cold water = crack).
- **2. Inspection:**
 - Check for shading (e.g., new trees, buildings) that might be blocking the sun.
 - Check that all wires are still secure and not chewed by rodents.
 - Check for any cracks in the panel glass after a hailstorm.

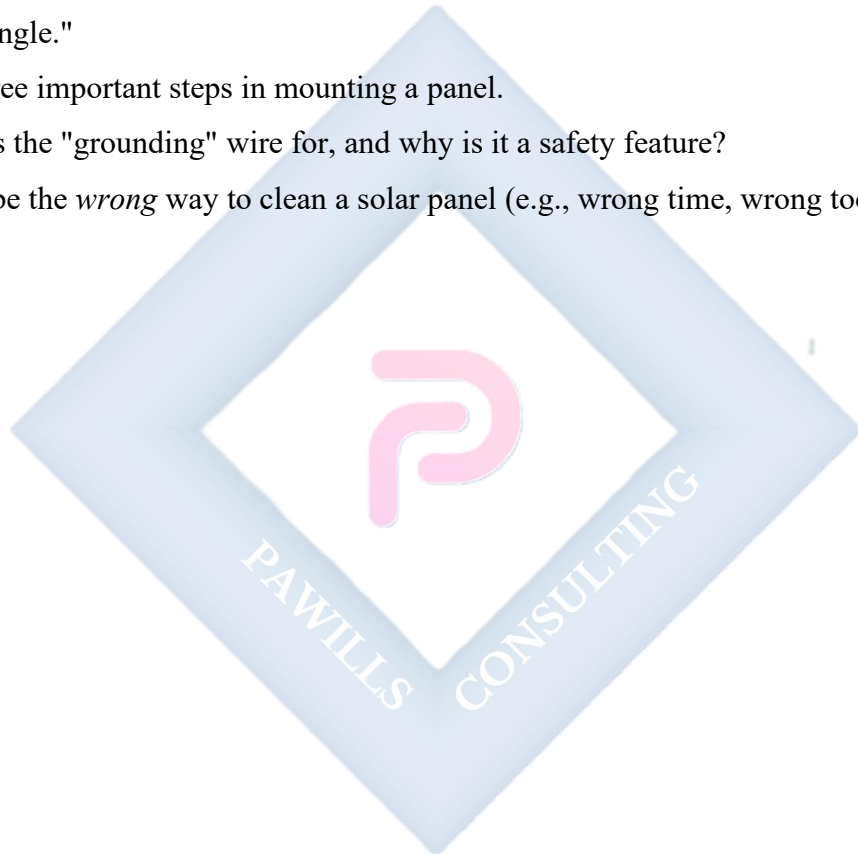
Evaluation:

1. What is the visual difference between a "mono" and a "poly" panel?
2. In Nigeria, what *direction* must a solar panel face?
3. What is the best *tilt angle* for a panel in Nigeria?

4. Why is it important to clean solar panels?
5. When is the best time of day to clean a panel, and why?

Assignment:

1. Why is a monocrystalline panel more expensive than a polycrystalline one?
2. Draw a simple diagram showing a panel on a roof, labeling the "South" direction and the "Tilt Angle."
3. List three important steps in mounting a panel.
4. What is the "grounding" wire for, and why is it a safety feature?
5. Describe the *wrong* way to clean a solar panel (e.g., wrong time, wrong tools).



WEEK 9: ELECTRICAL WIRING IN SOLAR SYSTEMS

Topic: Electrical Wiring in Solar Systems

Breakdown (Subtopics):

- Types of electrical wiring
- Wiring tools and materials
- Cable sizing and color codes
- Practical: wiring a small solar-powered circuit



Performance Objectives: By the end of this lesson, students should be able to:

1. List five essential wiring tools.
2. Define "cable sizing" and explain its importance for safety.
3. State the standard color codes for AC wiring (Live, Neutral, Earth).
4. State the standard color codes for DC wiring (Positive, Negative).
5. Describe the steps for wiring a simple solar circuit.

Lesson Content:

1. Wiring Tools and Materials

- **Tools:**
 - **Wire Stripper:** Used to remove the plastic insulation from a wire without cutting the copper inside.
 - **Wire Cutter:** For cutting wires to the correct length.
 - **Crimping Tool:** Used to attach "lugs" or "connectors" to the end of thick cables.
 - **Multimeter:** (Review) Used to check voltage, current, and continuity (to see if a wire is broken).
 - **Screwdrivers & Spanners:** For tightening terminals.
- **Materials:** Wires (Cables), Lugs, Junction Boxes, Conduit (pipes for wires).

2. Cable Sizing (Very Important for Safety)

- **Definition:** "Cable sizing" is the process of choosing the correct **thickness** (cross-sectional area) for a wire based on the amount of **current (Amps)** it will carry.
- **Measurement:** Cable thickness is measured in mm^2 (e.g., 4mm^2 , 6mm^2 , 16mm^2).
- **Why it Matters (Safety):**
 - All wires have resistance (Ohm's Law).
 - When current flows, this resistance creates *heat*.
 - **High Current (Amps) + Thin Wire (High Resistance) = EXTREME HEAT.**
 - If you use a wire that is *too thin* for the current, it will overheat, melt the plastic insulation, and **cause a fire**.
- **The Rule:**
 - **High Amps = Thick Cable** (e.g., Battery to Inverter)

- **Low Amps = Thin Cable** (e.g., Panel to Controller)

3. Standard Electrical Color Codes You *must* follow color codes so that anyone (including you) can identify wires safely.

- **AC Wiring (From Inverter to Plugs/Appliances):**
 - **L = Live** (the "hot" wire): **BROWN** (or sometimes Red, Black)
 - **N = Neutral:** **BLUE**
 - **E = Earth** (Ground): **GREEN/YELLOW stripes**
- **DC Wiring (Panels and Batteries):**
 - **+ = Positive:** **RED**
 - **- = Negative:** **BLACK**

4. Practical: Wiring a Small Solar Circuit (Teacher demonstrates the connections, emphasizing safety)

- **Circuit:** 1 Panel → 1 Charge Controller → 1 Battery → 1 DC Bulb
- **Steps:**
 1. **Safety:** Ensure the solar panel is covered (to stop it from producing power).
 2. **Controller to Battery:**
 - Take a **RED** wire and connect the BAT + on the controller to the + terminal of the battery.
 - Take a **BLACK** wire and connect the BAT - on the controller to the - terminal of the battery.
 - (The controller lights up).
 3. **Panel to Controller:**
 - Uncover the panel. Use a multimeter to find the + and - wires.
 - Connect the + wire from the panel to the PV + on the controller.
 - Connect the - wire from the panel to the PV - on the controller.
 - (The controller shows the panel is charging).
 4. **Load to Controller:**
 - Connect the DC bulb's **RED** wire to the LOAD + on the controller.

- Connect the DC bulb's **BLACK** wire to the LOAD - on the controller.
- (The bulb should light up).

Evaluation:

1. What is a "wire stripper" used for?
2. What is "cable sizing"?
3. What happens if you use a wire that is too thin for the current?
4. In AC wiring, what are the colors for Live, Neutral, and Earth?
5. In DC wiring, what are the colors for Positive and Negative?

Assignment:

1. Why is cable sizing a critical *safety* issue?
2. List 4 tools needed for wiring.
3. Draw a simple wiring diagram for a 1-panel, 1-controller, 1-battery system. Use RED and BLACK lines to show the correct polarity.
4. What is the Golden Rule for connecting a charge controller?
5. What is the color code for the Earth (ground) wire? What is its purpose? (Safety).

WEEK 10: ENTREPRENEURSHIP IN SOLAR PV

Topic: Entrepreneurship in Solar PV

Breakdown (Subtopics):

- Business opportunities in solar PV
- Customer service and professionalism
- Solar PV market trends
- Simple marketing plan for solar business

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Entrepreneurship" in the context of solar energy.
2. List and describe three different business opportunities in the solar PV sector.
3. Define "customer service" and list three qualities of a professional installer.
4. Outline a simple 4-point marketing plan.

Lesson Content:

1. Entrepreneurship in Solar PV (Recap)

- **Definition:** An entrepreneur is a person who identifies a problem, creates a business (a solution), and takes a financial risk to make a profit.
- **The Problem:** Unreliable and expensive public electricity; generator noise and pollution.
- **The Solution (Business):** Providing clean, reliable solar power.

2. Business Opportunities in Solar PV The solar industry is a "value chain" with many job opportunities:

1. Installation Technician (The Installer):

- This is the *skilled labor*. The person who designs the system, mounts the panels, and does all the wiring. This requires technical training (like this class).

2. Sales and Distribution (The Seller):

- This person *sells* the components (panels, batteries, inverters). They may run a shop or an e-commerce website. This requires business skill.

3. Maintenance Services (The "Doctor"):

- This entrepreneur offers "Solar Maintenance" packages. They visit clients every month to clean the panels, check the batteries, and ensure the system is working well. This provides steady, recurring income.

4. Consulting and System Sizing (The "Architect"):

- This is an expert who visits a client's house, calculates their energy needs (an "energy audit"), and designs the *perfect* system (e.g., "You need 6 panels, 1 MPPT controller, and 4 batteries."). They sell the *design*, not the products.

3. Customer Service and Professionalism A good entrepreneur is not just a good technician; they are good with *people*.

- **Customer Service:** The support you give to your customers *before, during, and after* the purchase.
- **Qualities of a Professional Installer:**
 1. **Honesty (Most Important):** Do not lie to the customer. Do not sell them fake products. Do not promise that a small system can power a freezer and an AC. Be truthful. A good reputation brings more customers.
 2. **Neatness and Safety:** Keep your work area clean. Do not leave cut wires on the floor. Install all wires neatly in pipes (conduit). A neat job looks professional and *is* safer.
 3. **Communication:** Explain to the customer *how* their new system works. Show them how to turn it on/off and how to read the inverter.
 4. **Punctuality:** Show up on the day and at the time you promised.
 5. **After-Sales Support:** Answer your phone if the customer calls with a problem a week later.

4. Solar PV Market Trends

- **Growing:** The market in Nigeria is growing *very* fast.
- **Falling Prices:** The cost of solar panels and lithium batteries is decreasing.
- **New Technology:** More efficient hybrid inverters and MPPT controllers are becoming standard.
- **Government Interest:** The government is encouraging renewable energy.

5. Simple Marketing Plan (How to get customers)

1. **Business Name:** e.g., "JSS2 Smart Solar Solutions"
2. **Service:** What do you do? e.g., "Solar and Inverter Installation and Maintenance."
3. **Target Audience:** Who are your customers? e.g., "Homes, small businesses (shops, barbers), and small schools."
4. **Marketing Tools (How to reach them):**
 - **Word-of-Mouth:** Do a *perfect* job for your first client (e.g., your uncle). They will tell their friends. This is the most powerful marketing.
 - **Online:** Post pictures of your neat installations on WhatsApp Status, Facebook, and Instagram.
 - **Signage:** Have a small, professional sign on your van or at your workshop.

Evaluation:

1. What is a solar "entrepreneur"?
2. List three different business opportunities in the solar PV sector.
3. What is the *most important* quality of a professional installer? (Honesty).
4. Why is customer service important?
5. What is one way you can market your new solar business?

Assignment:

1. Define "Entrepreneurship."
2. Choose one of the business opportunities (e.g., Installer, Seller, Maintenance) and explain what that person does.
3. Why is it "bad business" to lie to a customer about what their solar system can power?

4. Create a simple 4-point marketing plan for your *own* imaginary solar business.
5. What is the "target audience" for most solar installers in Nigeria?



WEEK 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Comprehensive review of all topics covered

Performance Objectives: By the end of this lesson, students should be able to:

1. Diagram the four main components of a standalone system.
2. Differentiate AC/DC, PWM/MPPT, and Series/Parallel.
3. Recall the key safety rules for wiring and maintenance.
4. Discuss the business and environmental importance of solar PV.

Lesson Content:

1. Summary of Term 1 Topics (The "Solar Foundation" Term):

- **Week 1 (Review):** A solar system has 4 parts: **Panel** (makes DC), **Controller** (protects battery), **Battery** (stores DC), **Inverter** (converts DC to AC). Safety is critical.
- **Week 2 (Electricity):** We learned the fundamentals.
 - **Conductor/Insulator:** (Copper/Plastic).
 - **AC/DC:** (NEPA/Battery).
 - **$V=I \times R$ (Ohm's Law):** (Pressure = Flow $\times$ Resistance).
 - **Multimeter:** (Measures V, I, R).
- **Week 3 (Energy):**
 - **Renewable (Clean):** Solar, Wind, Hydro (Infinite).
 - **Non-Renewable (Dirty):** Petrol, Gas, Coal (Finite).
 - Clean energy stops **CO2** and **Global Warming**.
- **Week 4 (Configuration):**
 - **Standalone (Off-Grid):** Needs batteries, no NEPA.
 - **Hybrid:** Uses Solar + Battery + NEPA. (Best for Nigeria).
 - **Series:** V adds up. **Parallel:** I (Amps) adds up.

- **Week 6 (Controllers):** The "brain" that protects the battery from **overcharging** and **over-discharging**.
 - **PWM:** Cheaper, less efficient.
 - **MPPT:** Smarter, more efficient (gets up to 30% more power).
 - **Safety Rule:** Connect **BATTERY FIRST**.
- **Week 8 (Mounting):**
 - **Direction:** Must face **TRUE SOUTH**.
 - **Angle:** Must be **10-15 degrees**.
 - **Maintenance:** Clean panels (remove dust) in the early morning.
- **Week 9 (Wiring):**
 - **Sizing:** Thin wire + High Amps = **FIRE**. Always use thick wires for batteries.
 - **Color Code (DC):** + is Red, - is Black.
 - **Color Code (AC):** Live is Brown, Neutral is Blue, Earth is Green/Yellow.
- **Week 10 (Business):** Solar is a great business (high demand). An entrepreneur must be **honest** and **professional** to get more customers.

2. Class Activity: Q&A Review Session (Teacher asks rapid-fire questions to test recall of all key concepts).

- "What does an inverter do?" (DC to AC)
- "What does a controller protect?" (The Battery)
- "Which direction do we face panels in Nigeria?" (South)
- "What adds up in a series connection?" (Voltage)
- "What is the color for a DC Negative wire?" (Black)
- "What is safer: PWM or MPPT?" (Both are safe, MPPT is more *efficient*).

Evaluation: This week's evaluation is the First Term Examination.

Assignment:

1. Study all notes from Week 1 to Week 10 for the examination.
2. Draw a complete, labeled diagram of a standalone solar PV system (all 4 components).
3. Draw a diagram of two panels in *series* and two panels in *parallel*.

4. List 5 critical safety rules (from any week).
5. Explain why you must *connect the battery first* when wiring a charge controller.



SECOND TERM - JSS 2 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND MAINTENANCE LESSON NOTES

WEEK 1: REVIEW OF FIRST TERM WORK

Topic: Review of First Term Work

Breakdown (Subtopics):

- Recap of solar PV components and configurations
- Review of electrical wiring concepts
- Discussion of charge controller and battery connections
- Assessment of students' understanding through class discussions



Performance Objectives: By the end of this lesson, students should be able to:

1. Recall and list the four primary components of a standalone solar system and their functions.
2. Differentiate between AC and DC, and state the sources for each in a PV system.
3. Explain the difference between a series and a parallel connection.
4. State the "Golden Rule" of wiring a charge controller.
5. Verbally participate in discussions, demonstrating their understanding of Term 1 concepts.

Lesson Content:

1. Recap of Solar PV Components and Configurations

- **A. Components:**
 - **Solar Panels:** Convert sunlight (photons) directly into **DC electricity**. They are the "generator" of the system.
 - **Charge Controller:** The "brain" that protects the battery. It prevents **overcharging** (from the panels) and **over-discharging** (from the loads).
 - **Battery:** The "storage tank." It stores the DC energy from the panels for use at night or on cloudy days.
 - **Inverter:** The "converter." It changes **DC power** (from the battery) into **AC power** (for home appliances).
- **B. Configurations:**
 - **Standalone (Off-Grid):** A self-contained system with batteries, completely separate from NEPA. Ideal for remote areas.
 - **Grid-Tied (On-Grid):** A system with no batteries, connected to NEPA. It works *with* the grid but shuts down when NEPA is off.
 - **Hybrid:** The best of both. It has batteries *and* is connected to the grid (or a generator). It uses solar first, then batteries, then the grid as a final backup. This is the most common and practical system in Nigeria.

2. Review of Electrical Wiring Concepts

- **A. AC vs. DC:**
 - **DC (Direct Current):** Flows in one direction. Source: **Solar Panels, Batteries.**
 - **AC (Alternating Current):** Flows back and forth. Source: **NEPA, Generators, Inverters.**
- **B. Series vs. Parallel Connections:**
 - **Series (Stacking):** Connecting (+) to (-). **Voltage adds up (V+V).** Current (Amps) stays the same. (e.g., two 12V panels become 24V).
 - **Parallel (Sharing):** Connecting (+) to (+) and (-) to (-). **Current adds up (A+A).** Voltage stays the same. (e.g., two 100A batteries become 200A).
- **C. Safety & Sizing:**
 - **Ohm's Law ($V=IR$):** The relationship between Voltage (pressure), Current (flow), and Resistance (blockage).
 - **Cable Sizing:** Using a wire that is *too thin* for the current (Amps) will cause it to overheat and **start a fire**. High Amps *must* have thick cables (especially from the battery to the inverter).
- **D. Color Codes:**
 - **DC:** Red is Positive (+), Black is Negative (-).
 - **AC:** Brown is Live (L), Blue is Neutral (N), Green/Yellow is Earth (E).

3. Charge Controller and Battery Connections (The Golden Rule)

- **Rule:** When installing a charge controller, you **MUST CONNECT THE BATTERY FIRST!**
- **Reason:** The controller gets its power and "learns" the system voltage (12V? 24V?) from the battery. Connecting the high-voltage panel first can confuse or destroy the controller.
- **Connection Order:**
 1. **BATTERY** to Controller.
 2. **PANEL** to Controller.
 3. **LOAD** to Controller (if using the DC load terminals).
- **Disconnection Order:** (The exact reverse)

1. Disconnect **PANEL**.
2. Disconnect **LOAD**.
3. Disconnect **BATTERY**.

4. Class Discussion (Assessment) (Teacher leads a Q&A session)

- "What does an inverter do?"
- "Why is a hybrid system best for Nigeria?"
- "What happens to the voltage if I connect two 12V batteries in series?"
- "What happens if I use a 4mm wire for a 50A current?"
- "What is the first thing you must connect to your charge controller?"

Evaluation: (This is an oral evaluation based on the class discussion)

1. Explain the function of each of the four main components.
2. What is the difference between AC and DC?
3. What is the difference between a series and a parallel connection?
4. What is the main job of a charge controller?
5. What are two safety rules for wiring?

Assignment:

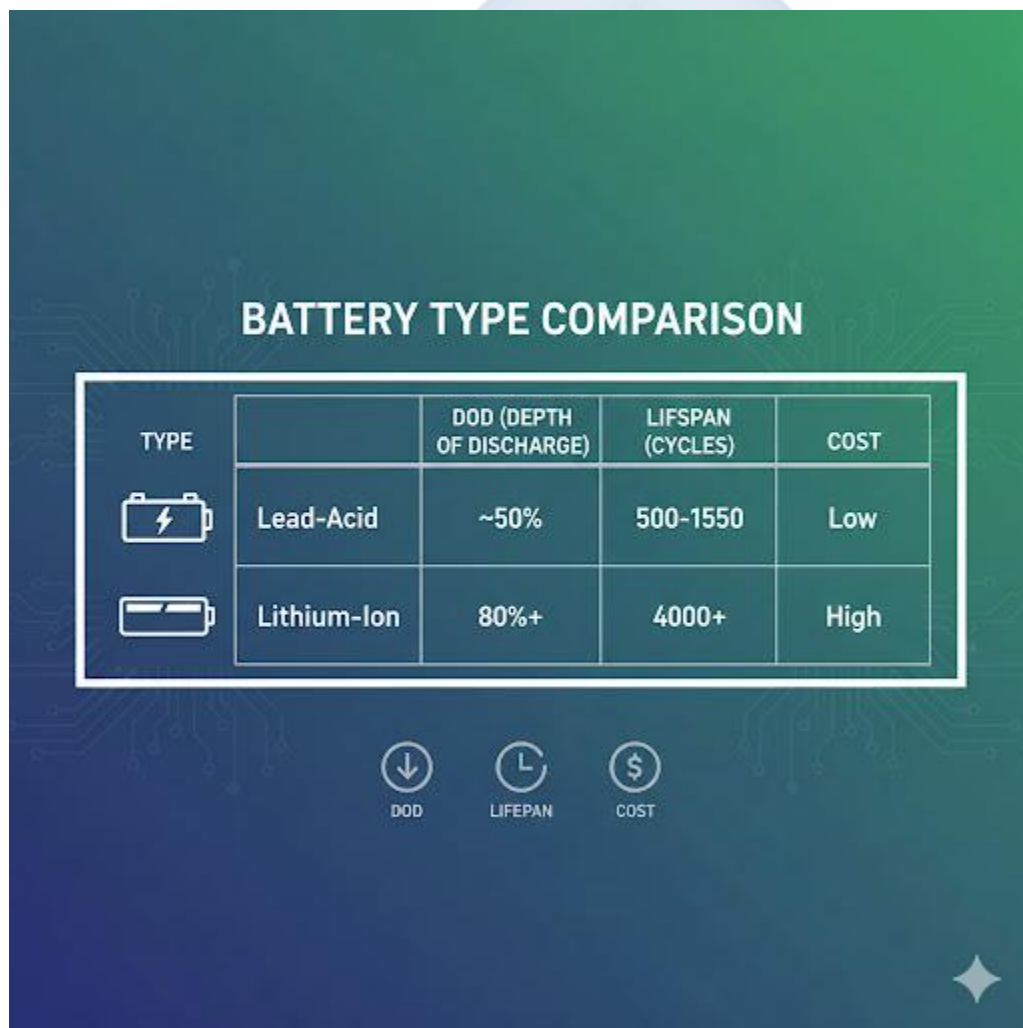
1. Draw a complete, labeled diagram of a **Hybrid Solar System**, showing all 5 components (Panel, Controller, Battery, Inverter, and the Grid/NEPA).
2. Why is a "grid-tied" system not very useful in Nigeria?
3. You have two 12V, 200Ah batteries.
 - (a) What is the total Voltage and Capacity (Ah) if you wire them in **series**?
 - (b) What is the total Voltage and Capacity (Ah) if you wire them in **parallel**?
4. Why is cable sizing so important for preventing fires?
5. What is the correct *order* for *disconnecting* all components from a charge controller?

WEEK 2: SOLAR BATTERIES

Topic: Solar Batteries

Breakdown (Subtopics):

- Meaning and importance of solar batteries
- Types of batteries: Lead-acid, lithium-ion, gel batteries
- Functions and characteristics of each type
- Battery charging and discharging cycles
- Practical: Identifying and testing batteries

A graphic titled "BATTERY TYPE COMPARISON" on a green and blue background with circuit patterns. It features a table comparing Lead-Acid and Lithium-Ion batteries. Below the table are three circular icons: a downward arrow for DOD, an 'L' for Lifespan, and a dollar sign for Cost. A small star icon is in the bottom right corner.

TYPE		DOD (DEPTH OF DISCHARGE)	LIFESPAN (CYCLES)	COST
	Lead-Acid	~50%	500-1550	Low
	Lithium-Ion	80%+	4000+	High

DOD

LIFESPAN

COST

Performance Objectives: By the end of this lesson, students should be able to:

1. State the primary importance of batteries in a solar system.
2. Differentiate between the three main types of solar batteries (Lead-Acid, Gel, Lithium-ion).
3. List two pros and two cons for each battery type.
4. Define "Depth of Discharge (DoD)" and "Battery Cycle."
5. Use a multimeter to test a battery's DC voltage.

Lesson Content:

1. Meaning and Importance of Solar Batteries

- **Meaning:** A solar battery is a device that stores chemical energy and converts it into electrical energy (DC) on demand.
- **Importance (The "Storage Tank"):** Solar panels only produce power during the day. A battery is essential in an off-grid or hybrid system to **store this excess energy** for use at night or on cloudy/rainy days. Without a battery, you would have no power as soon as the sun goes down.

2. Types of Solar Batteries

- **A. Lead-Acid Batteries (Deep Cycle):**
 - **How they work:** The oldest technology. They consist of lead plates hanging in a bath of sulphuric acid and water.
 - **Sub-type: Flooded Lead-Acid (FLA):** These are the traditional "wet" batteries.
 - *Pros:* Cheapest option.
 - *Cons:* Require regular **maintenance** (must top up with distilled water), must be kept upright, release explosive gases (need good ventilation).
 - **Sub-type: Sealed Lead-Acid (SLA) / VRLA:**
 - **Gel Batteries:** The acid is mixed into a gel, so it can't spill.
 - **AGM (Absorbent Glass Mat):** The acid is held in a fibreglass mat.
 - *Pros:* **Maintenance-free** (no watering), sealed, and safer.

- **Cons:** More expensive than FLA, can be damaged by overcharging.

- **B. Lithium-ion (LiFePO4):**

- **How they work:** The newest technology, similar to what's in your phone or laptop, but larger.
- **Pros:**

1. **Best Performance:** Can be discharged much deeper.
2. **Longest Lifespan:** Lasts 5-10 times longer than lead-acid.
3. **Lightweight & Compact:** Much smaller and lighter for the same amount of power.
4. **Maintenance-Free.**
 - **Cons:** **Very high initial cost** (but often cheaper in the long run because you don't replace it).

3. Battery Characteristics (Key Terms)

- **A. Battery Cycle (Lifespan):**

- **Definition:** One "cycle" is one full charge and one discharge.
- **Lifespan:** Batteries are rated by how many *cycles* they can perform before they die.
 - **Lead-Acid:** 500 - 1,500 cycles (2-4 years of use)
 - **Lithium-ion:** 4,000 - 8,000 cycles (10-20 years of use)

- **B. Depth of Discharge (DoD):**

- **Definition:** This is the *percentage* of the battery's total energy that you can safely *use* before you must recharge it.
- **Lead-Acid:** Has a **50% DoD**. If you have a 200Ah battery, you should only ever use 100Ah before recharging. Using more will *permanently damage* it.
- **Lithium-ion:** Has an **80-100% DoD**. You can use almost all the energy.
- **Importance:** A 200Ah Lithium battery has almost *double* the usable energy of a 200Ah Lead-Acid battery.

4. Practical: Identifying and Testing Batteries

- **Identifying:** Read the label on the battery. It will state:

- **Voltage:** 12V (most common)
- **Capacity:** 100Ah, 150Ah, 200Ah (Amp-Hours)
- **Type:** Flooded, GEL, or LiFePO4

- **Testing (with a Multimeter):**

1. Set the multimeter dial to **DC Voltage (V⁻)**.
2. Touch the **Red** probe to the **Positive (+)** terminal.
3. Touch the **Black** probe to the **Negative (-)** terminal.
4. Read the voltage:
 - A full 12V battery will read **~12.7V - 13.0V**.
 - A half-full battery will read **~12.2V**.
 - An "empty" (fully discharged) battery will read **~11.8V or less**.

Evaluation:

1. Why is a battery essential for a hybrid or standalone solar system?
2. What is the main *disadvantage* of a "Flooded" Lead-Acid battery?
3. What is the main *advantage* of a "Lithium-ion" battery?
4. What is "Depth of Discharge (DoD)"?
5. What is the DoD of a Lead-Acid battery? What about a Lithium-ion battery?

Assignment:

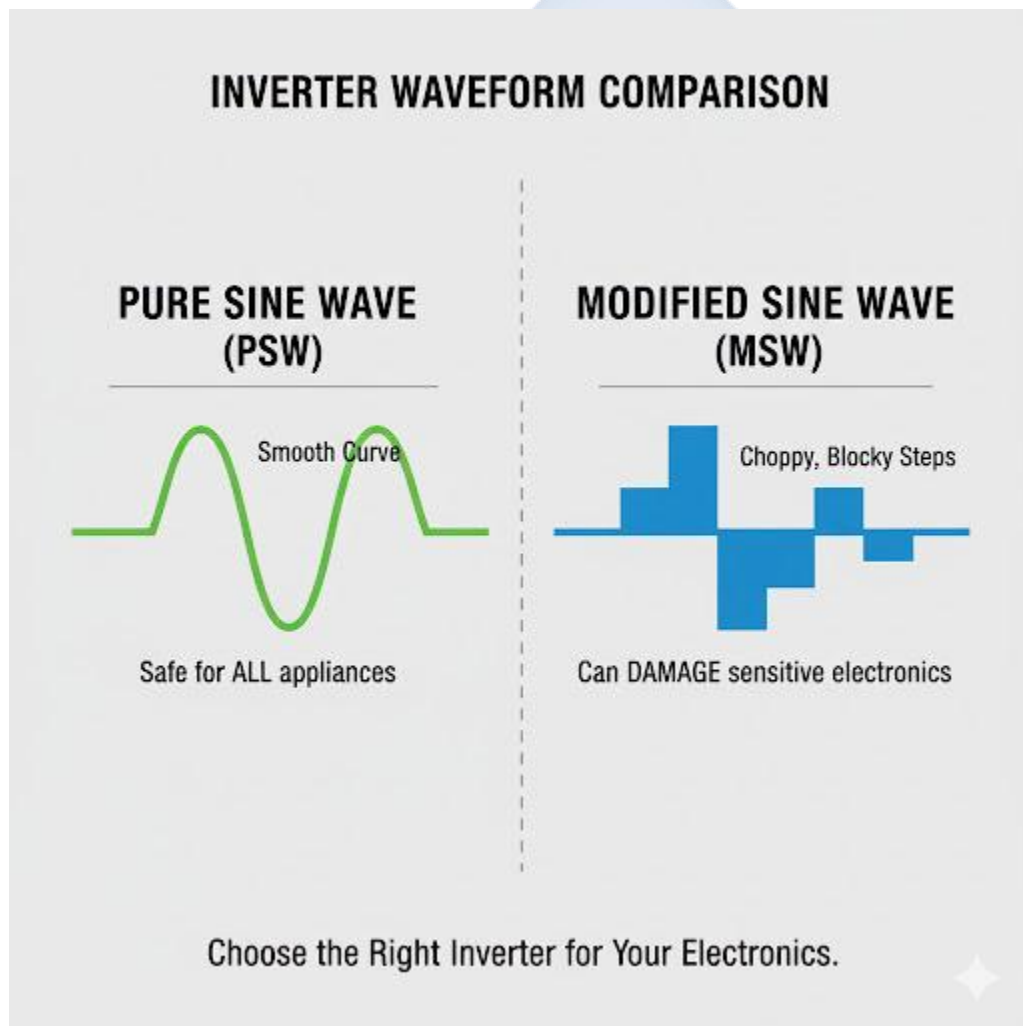
1. Define "Battery Cycle" and "Depth of Discharge."
2. Make a table with three columns: "Battery Type," "Pros," "Cons."
3. Fill in the table for Flooded Lead-Acid, Gel, and Lithium-ion.
4. You have a 12V, 200Ah Lead-Acid battery (50% DoD) and a 12V, 200Ah Lithium-ion battery (90% DoD). How many *usable* Amp-Hours (Ah) does each battery have?
5. Describe how to use a multimeter to check if a 12V battery is full or empty.

WEEK 3: INVERTERS AND CONVERTERS

Topic: Inverters and Converters

Breakdown (Subtopics):

- Meaning and functions of inverters and converters
- Types of inverters: pure sine wave, modified sine wave
- Importance of voltage conversion (DC to AC)
- Practical: Connecting an inverter to a solar battery



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Inverter" and "Converter."

2. State the primary function of an inverter (DC to AC).
3. Differentiate between a Pure Sine Wave (PSW) and a Modified Sine Wave (MSW) inverter.
4. Explain which type of inverter is safe for all appliances.
5. Safely connect an inverter to a battery.

Lesson Content:

1. Meaning and Functions

- **Inverter:**
 - **Function:** An inverter is an electronic device that **converts Direct Current (DC) power into Alternating Current (AC) power.**
 - **Why it's essential:** Your solar panels and batteries supply low-voltage DC (e.g., 12V). Your home appliances (TV, fans, fridge, sockets) run on high-voltage AC (e.g., 230V). The inverter "inverts" the power to make it usable.
- **Converter:**
 - **Function:** This is a more general term. An inverter is one type of converter (DC-to-AC).
 - Other types include:
 - **Rectifier:** Converts AC to DC (e.g., your phone charger).
 - **DC-to-DC Converter:** Converts one DC voltage to another (e.g., 12V from a battery down to 5V to charge a phone via USB).

2. Types of Inverters This is the *most important* feature to check when buying an inverter. It describes the *quality* of the AC electricity it produces.

- **A. Modified Sine Wave (MSW) Inverter:**
 - **Definition:** An older, cheaper type of inverter.
 - **Waveform:** It produces a "choppy" or "blocky" AC signal. It is not a clean, smooth wave.
 - **Use:** It can safely power *simple* "resistive" loads like old-fashioned bulbs or heaters.

- **DANGER:** It can **damage** or **destroy** "sensitive" or "inductive" electronics.
 - **Do NOT use with:** Fans (will hum), fridges, TVs, laptops, computers, or any appliance with a motor.
- **Analogy:** It's like "dirty" or "low-quality" fuel for your appliances.
- **B. Pure Sine Wave (PSW) Inverter:**
 - **Definition:** A modern, high-quality, (and more expensive) inverter.
 - **Waveform:** It produces a perfectly smooth, clean, alternating wave, *identical* to (or even better than) the power from NEPA.
 - **Use: Safe for ALL appliances.** This is the *only* type of inverter you should use for a home or business. It can power fans, fridges, TVs, and sensitive laptops without any risk.
 - **Analogy:** It's like "premium" or "clean" fuel for your appliances.

3. Importance of Voltage Conversion (Summary) The inverter is the "gateway" between your solar system and your home.

- It takes **low-voltage DC** (e.g., 12V, 24V) from the batteries.
- It converts it to **high-voltage AC** (e.g., 230V, 50Hz) for your sockets.
- The *quality* of this conversion (PSW vs. MSW) determines which appliances you can safely use.

4. Practical: Connecting an Inverter to a Battery

- **SAFETY FIRST:** Inverters handle high currents and can produce high (dangerous) voltage.
 1. **Turn Inverter OFF:** Ensure the inverter's power switch is in the "OFF" position.
 2. **Get Correct Cables:** You must use *very thick* cables (as specified by the manufacturer) for the battery-to-inverter connection. Thin cables will **melt or catch fire**.
 3. **Connect Negative (BLACK):** Connect the **BLACK** cable to the **Negative (-)** terminal of the inverter. Then connect the other end to the **Negative (-)** terminal of the battery.

4. **Connect Positive (RED):** Connect the **RED** cable to the **Positive (+)** terminal of the inverter.
5. **Connect to Battery (SPARK):** Briefly tap the **RED** cable lug to the **Positive (+)** terminal of the battery. You will see a **small, sharp spark**. This is *normal*. It is the inverter's capacitors charging.
6. **Secure Connection:** Firmly bolt the RED cable to the Positive (+) battery terminal.
7. **Turn Inverter ON:** The inverter should now power on.
8. **Test:** Plug a simple AC appliance (like a bulb) into the inverter's socket to test.

Evaluation:

1. What is the main function of an inverter?
2. What is the main function of a rectifier (like your phone charger)?
3. What is the difference between a Pure Sine Wave (PSW) and Modified Sine Wave (MSW) inverter?
4. Which type of inverter is safe for *all* appliances, including TVs and fridges?
5. What should you expect to see when connecting the *second* (positive) inverter cable to the battery?

Assignment:

1. Define Inverter and Converter.
2. Draw a simple diagram showing a battery connected to an inverter, which is then connected to a TV. Label the "DC" part and the "AC" part of your diagram.
3. Why should you *not* use a Modified Sine Wave (MSW) inverter to power a fan or a laptop?
4. What is the "safest" type of inverter, and why?
5. What is the first step you must take *before* connecting an inverter to a battery? (Ensure it is switched OFF).

WEEK 4: LOAD ESTIMATION AND ENERGY DEMAND

Topic: Load Estimation and Energy Demand

Breakdown (Subtopics):

- Meaning of electrical load
- Steps in calculating total load requirement
- Daily energy consumption estimation
- Importance of proper load management
- Practical: Calculating the load of classroom equipment

Appliance	Power(W)	Time (h)	Energy (Wh)
LED Bulb	10	6	60
Ceiling Fan	70	8	560
Television	100	5	500
TOTAL	180	-	1120

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Electrical Load," "Power" (Watts), and "Energy" (Watt-hours).
2. Differentiate between Power and Energy.
3. List the steps required to conduct a "load audit."
4. Calculate the total daily energy demand (in Watt-hours) for a simple list of appliances.
5. Explain the importance of load management.

Lesson Content:

1. Meaning of Electrical Load, Power, and Energy

- **Load:** An electrical load is any appliance or device that *consumes* electricity to do work (e.g., a bulb, fan, TV, fridge).
- **Power (Watts):** This is the *rate* at which an appliance uses energy. It is how "power-hungry" the device is.
 - **Unit: Watts (W).**
 - *Example:* A small bulb might be **10W**. A large fridge might be **150W**.
- **Energy (Watt-hours):** This is the *total amount* of electricity used over a *period of time*.
 - **Unit: Watt-hours (Wh).**
 - **Formula: Energy (Wh) = Power (W) × Time (h)**
 - *Example:* A 10W bulb left on for 5 hours uses: $10\text{W} \times 5\text{h} = \mathbf{50\ Wh}$ of energy.
 - A 150W fridge that runs for 8 hours a day uses: $150\text{W} \times 8\text{h} = \mathbf{1,200\ Wh}$ of energy.

2. Steps in Calculating Total Load (Load Audit) This is the *first* and *most important* step in designing *any* solar system.

- **Step 1: LIST all appliances** you want to power. (e.g., 4 bulbs, 1 TV, 2 fans, 1 fridge).
- **Step 2: FIND the Power (W) of each appliance.** (Check the sticker on the back of the appliance).
- **Step 3: ESTIMATE the Time (h)** you will use *each* appliance per day. (Be realistic).
- **Step 4: CALCULATE the Energy (Wh)** for each appliance. ($\text{W} \times \text{h} = \text{Wh}$).
- **Step 5: ADD up the Watt-hours** to get your Total Daily Energy Demand.

3. Practical: Calculating the Load of Classroom Equipment

Let's do a load audit for our

classroom (example):

Appliance	Quantity	Power (Watts)	Hours of Use (per day)	Energy (Watt-hours)
LED Bulb	4	10 W	6 h	$4 \times 10\text{W} \times 6\text{h} = 240\ \text{Wh}$
Ceiling Fan	2	80 W	6 h	$2 \times 80\text{W} \times 6\text{h} = 960\ \text{Wh}$
Laptop	1	65 W	4 h	$1 \times 65\text{W} \times 4\text{h} = 260\ \text{Wh}$
TOTALS		Running Watts: 385 W		Daily Energy: 1,460 Wh

- **Running Watts (Power):** $40\text{W (bulbs)} + 160\text{W (fans)} + 65\text{W (laptop)} = \mathbf{265\text{ W}}$. (This is the *instant* power needed. The inverter must be *larger* than this).
- **Total Energy Demand:** $240\text{ Wh} + 960\text{ Wh} + 260\text{ Wh} = \mathbf{1,460\text{ Wh per day}}$. (This is the *total energy* needed. The batteries and panels must be sized to provide this amount).

4. Importance of Proper Load Management

- **You cannot power everything:** A solar system has a limited budget. You cannot run an AC, an electric cooker, or a water heater on a small system.
- **Load Management:** This means being smart about *what* you use and *when* you use it.
 - Use energy-efficient appliances (e.g., LED bulbs, not 100W incandescent bulbs).
 - Do not turn on all heavy loads at the same time (it will overload the inverter).
 - Iron clothes during the day (when the sun is shining) so the power comes from the panels, not the battery.

Evaluation:

1. What is an "electrical load"?
2. What is the difference between Power (Watts) and Energy (Watt-hours)?
3. What is the formula for calculating Energy?
4. List the 5 steps of a load audit.
5. A 50W fan runs for 10 hours. How much energy (in Wh) does it consume?

Assignment:

1. Define Power (Watts) and Energy (Watt-hours).
2. Why is a load audit the *first* step in designing a solar system?
3. What does "load management" mean? Give one example.
4. **Practical:** Do a simple load audit for your *own* room at home.
 - (a) List 3-4 appliances (bulb, fan, TV, phone charger).
 - (b) Find their Power (Watts) (you can estimate: Bulb=15W, Fan=70W, TV=100W, Charger=10W).
 - (c) Estimate your Hours of Use per night.

- (d) Calculate the **Total Daily Energy (Wh)** you need for your room.
5. What is the total *Running Watts* from your calculation in Q4? What size inverter (in Watts) would you need?

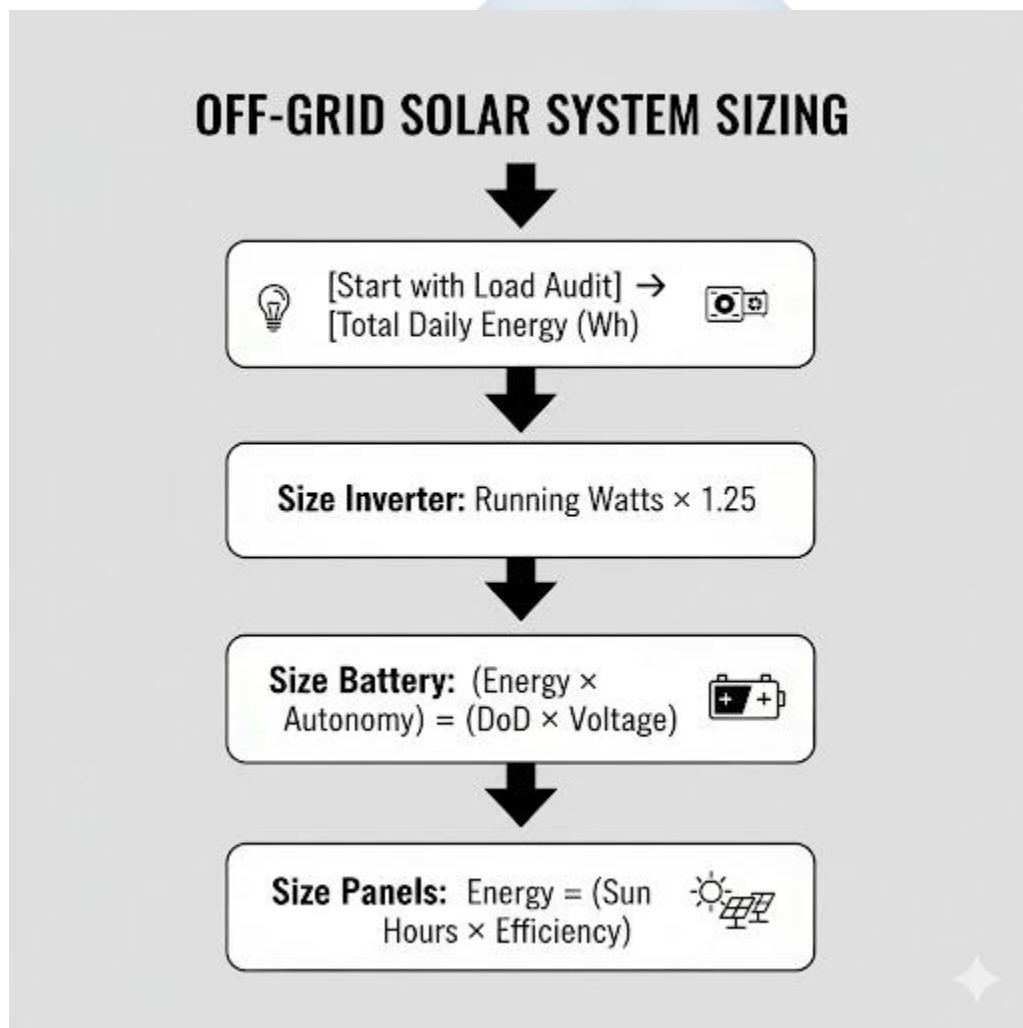


WEEK 6: SYSTEM SIZING AND DESIGN

Topic: System Sizing and Design

Breakdown (Subtopics):

- Meaning of system sizing
- Factors to consider in sizing (load, autonomy, location)
- Panel, battery, and inverter sizing
- Design of a small off-grid solar system
- Practical: Solar system design demonstration



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "System Sizing."
2. List the three key factors needed for sizing (Load, Autonomy, Sun Hours).
3. Calculate the required Inverter size based on a load.
4. Calculate the required Battery Bank size based on a load and autonomy.
5. Calculate the required Solar Panel array size based on a load and sun hours.

Lesson Content:

1. Meaning of System Sizing

- **Definition:** System Sizing is the process of using calculations to determine the **correct size and quantity** of solar components (Inverter, Batteries, Panels, Controller) needed to meet a specific electrical load.
- **Goal:** To design a system that is *reliable* (always has power) and *cost-effective* (not wasting money on components that are too big).

2. Factors to Consider in Sizing

1. **Total Load (Energy Demand):** (From Week 4). This is the *most important* factor. How many Watt-hours (Wh) do you use per day?
 - *Our Example:* 1,460 Wh/day (from Week 4 classroom load).
2. **Days of Autonomy (Battery):** How many days do you want your system to run *without any sun?* (e.g., during heavy rain/clouds).
 - *Standard:* 1 or 2 days. (Let's use 1 day).
3. **Peak Sun Hours (PSH) (Panels):** The average number of hours per day that the sun is at its *peak* strength. This is based on your *location*.
 - *Nigeria (Average):* 4.5 Peak Sun Hours (PSH).

3. Sizing Calculations (The Three Steps) Let's use our Classroom Load: **Total Running Watts = 265 W | Total Energy = 1,460 Wh/day**

- **Step 1: Size the INVERTER (based on Power, in Watts)**

- **Rule:** The inverter must be larger than the total *running* Watts. A safety margin of 25% is added.
- **Formula:** Inverter Size (W) = Total Running Watts $\times$ 1.25
- **Calculation:** 265 W $\times$ 1.25 = 331 W
- **Result:** You must buy the next available size up. **A 500W Inverter** (or 600W) would be perfect. (You also need a 12V inverter to match a 12V battery).
- **Step 2: Size the BATTERY BANK (based on Energy, in Wh)**
 - **Rule:** The battery must store your *total energy* + *autonomy*, respecting the DoD. (We will assume a 12V system and Lead-Acid 50% DoD).
 - **Formula:** Total Battery Capacity (Ah) = (Total Energy (Wh) $\times$ Days of Autonomy) / (DoD $\times$ System Voltage)
 - **Calculation:** (1,460 Wh $\times$ 1 day) / (0.50 DoD $\times$ 12V)
 - **Calculation:** 1,460 / 6 = 243.3 Ah
 - **Result:** You need **243.3 Ah** of *usable* capacity at 12V. Since a 200Ah battery is common, you would need **two** 200Ah batteries in parallel (for 400Ah total, giving 200Ah usable). Or, a single **250Ah** battery.
- **Step 3: Size the SOLAR PANELS (based on Energy, in Wh)**
 - **Rule:** The panels must replace 100% of the energy you use *every day*, accounting for system losses (e.g., heat, wire resistance. We'll use 0.77 efficiency).
 - **Formula:** Total Panel Watts = Total Energy (Wh) / (Peak Sun Hours $\times$ System Efficiency)
 - **Calculation:** 1,460 Wh / (4.5 PSH $\times$ 0.77)
 - **Calculation:** 1,460 / 3.465 = 421 W
 - **Result:** You need **421 Watts** of solar panels. You could buy **two** 250W panels (total 500W) or **three** 150W panels (total 450W).

4. Design Demonstration (Summary) For our classroom load (1,460 Wh/day), we need:

- **Inverter:** 1 x 500W (12V, Pure Sine Wave)
- **Battery:** 2 x 200Ah (12V) Lead-Acid batteries
- **Solar Panels:** 2 x 250W panels

- **Charge Controller:** A 40A MPPT controller (sized to handle the panel's current and voltage).

Evaluation:

1. What is "system sizing"?
2. What are the three main factors you must know *before* you can size a system?
3. Which component is sized based on *Running Watts*?
4. Which component is sized based on *Watt-hours* and *Days of Autonomy*?
5. Which component is sized based on *Watt-hours* and *Peak Sun Hours*?

Assignment:

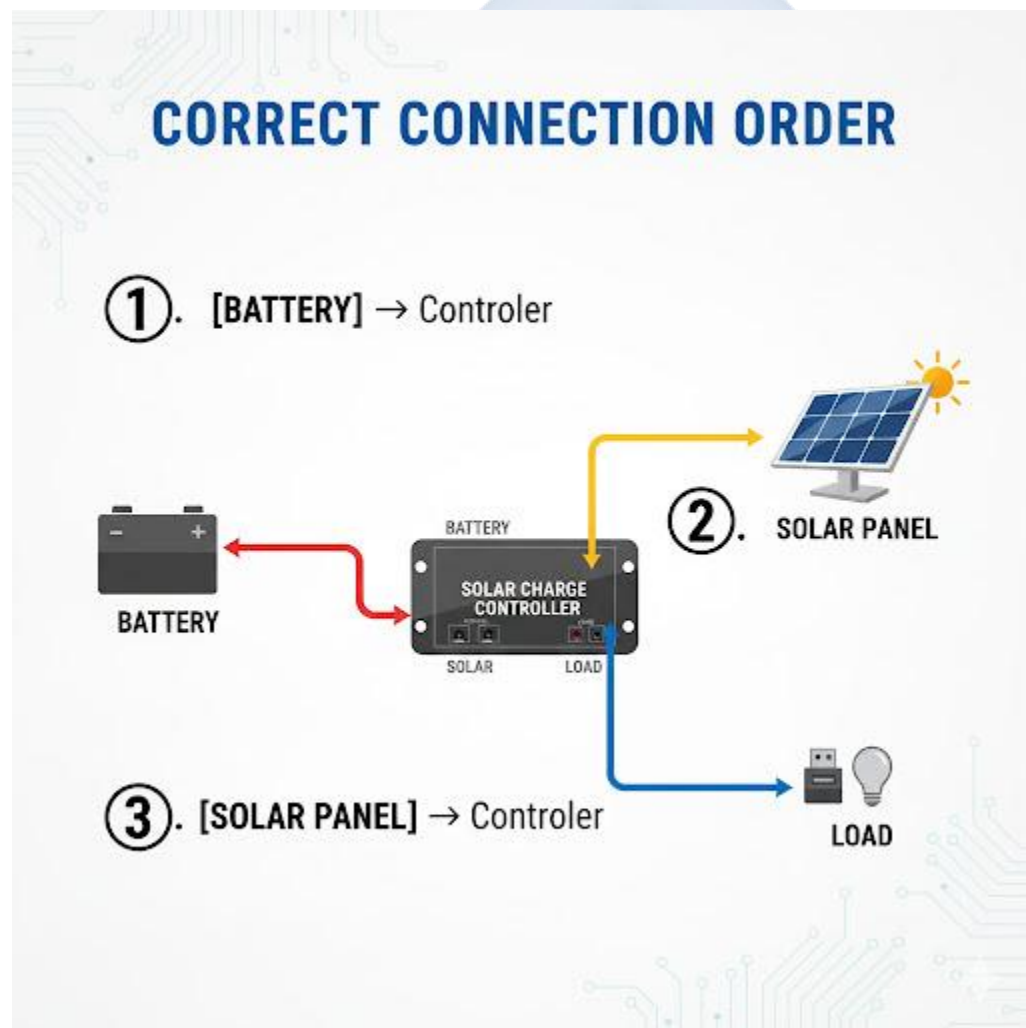
1. Define System Sizing, Days of Autonomy, and Peak Sun Hours.
2. Use your personal load audit from Week 4 (Assignment Q4).
3. **Calculate the Inverter size** you would need for *your* room. (Total Running Watts $\times$ 1.25).
4. **Calculate the Battery Bank size (Ah)** you would need. (Use 1 day autonomy, 50% DoD, and 12V).
5. **Calculate the Solar Panel size (W)** you would need. (Use 4.5 PSH and 0.77 efficiency).

WEEK 8: INSTALLATION PROCEDURES

Topic: Installation Procedures

Breakdown (Subtopics):

- Pre-installation site inspection
- Steps in solar PV installation
- Wiring connection and testing
- Common installation errors and troubleshooting
- Demonstration: Installing a small solar lighting system



Performance Objectives: By the end of this lesson, students should be able to:

1. List three tasks to perform during a pre-installation site inspection.
2. List the 5 main steps of a solar installation.
3. Identify two of the most common (and dangerous) installation errors.
4. Describe the "commissioning" or testing process.

Lesson Content:

1. Pre-Installation Site Inspection This is the *planning* you do at the client's house *before* you buy or install anything.

1. Roof Assessment:

- **Direction:** Find **True South** (using a compass app). Is there a roof that faces south?
 - **Shading:** Look for trees, other buildings, or satellite dishes that will cast a **shadow** on the roof during the day. A small shadow can kill the power of a whole panel.
 - **Condition:** Is the roof strong enough? Is it leaking?
2. **Cable Route:** Plan the path the wires will take from the roof (panels) to the "balance of system" (indoors).
 3. **BOS Location:** Find a cool, dry, ventilated, and secure indoor location (like a store or garage) to install the inverter, controller, and batteries. **Never** install batteries in a hot, sealed room or a living room (due to gas release).

2. Steps in Solar PV Installation (The "Big Picture")

1. **Step 1: Mount Racking & Panels:** (Term 1, Week 8) Secure the mounting brackets to the roof, build the rails, and bolt the panels down (facing South at 10-15°).
2. **Step 2: Install "Balance of System" (BOS):** Mount the Inverter, Charge Controller, and Breakers/Fuses on a wall indoors, near the battery box.

3. **Step 3: Run Wiring (The "Home Run"):** Run the DC cables (in conduit/pipe) from the panels on the roof down to the controller (indoors). Run the AC cables from the inverter to the house's distribution board ("fuse box").
4. **Step 4: Connect Components:** Wire all the components together. (See "Golden Rule" in Step 3).
5. **Step 5: Commissioning (Testing):** Turn on the system in the correct order and test all functions.

3. Wiring Connection and Testing (Commissioning)

- **Connection Order (The ONLY safe way):**
 1. Wire the Battery Bank (e.g., parallel/series).
 2. Connect **Battery to Inverter** (leave inverter OFF).
 3. Connect **Battery to Charge Controller** (controller lights up).
 4. Connect **Panel to Charge Controller** (controller shows charging).
- **Testing (Commissioning):**
 1. Turn on the Charge Controller (if it has a switch).
 2. Use a multimeter to check the voltage from the panels (PV) and the voltage of the battery (BAT) on the controller's terminals.
 3. Turn on the Inverter. Listen for a beep or hum.
 4. Plug a test load (e.g., a simple lamp) into the inverter's AC socket. It should light up.
 5. Turn on the main AC breaker to the house. The house now has solar power.

4. Common Installation Errors and Troubleshooting

- **Reversed Polarity (The #1 Error):** Connecting Positive (+) to Negative (-). This will *instantly* destroy the charge controller or inverter. It is an expensive mistake. **Always use a multimeter to check polarity before connecting.**
- **Loose Connections:** Terminals that are not screwed down tightly will create high resistance, get very hot, and **melt or cause a fire.**

- **Incorrect Cable Sizing:** Using thin wires for batteries or inverters. This is a major fire hazard.
- **Ignoring Shading:** Mounting a panel in a spot that gets a shadow at 3 p.m. This will cripple the system's performance every afternoon.

Evaluation:

1. What is a "site inspection"?
2. List the 5 main steps of a full installation.
3. What is "shading," and why is it a problem?
4. What is "reversed polarity"? What does it do?
5. What is the correct, safe order for connecting all the components?

Assignment:

1. Why is a pre-installation site inspection necessary? List two things you would check.
2. What is the most common and dangerous installation error? How can you prevent it?
3. What is a "loose connection," and why is it a fire hazard?
4. What does it mean to "commission" a system?
5. You install a system, but the batteries are not charging. List two possible faults (e.g., "panel wire is broken," "controller is damaged," "reversed polarity on panel").

WEEK 9: SAFETY PRACTICES IN INSTALLATION

Topic: Safety Practices in Installation

Breakdown (Subtopics):

- Importance of safety in solar installation
- Personal protective equipment (PPE)
- Safe handling of tools and materials
- Electrical hazard prevention
- First aid and emergency response



Performance Objectives: By the end of this lesson, students should be able to:

1. State the three main hazards in solar installation (electrical, chemical, physical).
2. List the essential Personal Protective Equipment (PPE) for an installer.
3. Describe the safe method for handling batteries.
4. Describe how to prevent DC arc flashes (short circuits).
5. State the first action to take in case of an electrical shock emergency.

Lesson Content:

1. Importance of Safety (The 3 Main Hazards) Solar installation is a dangerous job. You must protect yourself from three types of hazards:

1. **Electrical Hazards:** Shock and **Arc Flash** from DC power (panels/batteries) and AC power (inverter/grid). DC arcs are *worse* than AC arcs because they do not self-extinguish.
2. **Chemical Hazards:** Battery acid (sulphuric acid) can cause severe burns to skin and eyes. Batteries can also release explosive **hydrogen gas**.
3. **Physical Hazards:** Falling from a roof or ladder. Lifting heavy panels or batteries.

2. Personal Protective Equipment (PPE) This is the "uniform" of a professional installer.

- **Eyes: Safety Goggles.** *Mandatory* when working with batteries (to protect from acid splash) and when drilling.
- **Hands: Insulated Gloves** (rubber-coated) when making live connections. **Work Gloves** for handling panels and racks.
- **Feet: Rubber-Soled Boots** (or safety boots). These insulate you from the ground, preventing shock.
- **Body:** Long-sleeved shirts (to protect from sun and sparks).
- **Head: A Hard Hat** if working on a busy construction site or under other workers.

3. Safe Handling of Tools and Materials

- **Tools:** Use **insulated tools** (screwdrivers, spanners) with rubber handles. Never use tools with bare metal handles.
- **Batteries:**
 - **Gas:** ALWAYS charge batteries in a **well-ventilated** area (not a sealed room).
 - **Sparks:** NEVER smoke, weld, or create sparks near a battery. The hydrogen gas can ignite and cause a battery *explosion*.
 - **Shorting:** Never place a metal tool (like a spanner) on *top* of a battery. If it touches the (+) and (-) terminals, it will create a massive short circuit.
- **Panels:**
 - Cover the panel with a blanket or cardboard *before* wiring it. This stops it from producing dangerous DC voltage while you work.

4. Electrical Hazard Prevention

- **DC Arc Flash:** A DC short circuit (e.g., (+) and (-) wires touching) is extremely dangerous. It creates a "plasma" arc that is hotter than the sun and can cause blindness, severe burns, and fires.
- **Prevention:**
 1. **Always use Fuses and Breakers:** These are "sacrificial" safety devices. They will blow up *first* and cut the circuit before the wire melts or a fire starts.
 2. **One Hand Rule:** When working in a "live" box, try to work with only one hand at a time. This prevents electricity from flowing *across your heart* (e.g., from your left hand to your right hand).
 3. **Check Polarity:** Use your multimeter 3-4 times to check: Is this wire Red? Is it *really* positive?
 4. **Insulate Everything:** No bare wires should be exposed. Use electrical tape or connectors.

5. First Aid and Emergency Response

- **If someone gets an electric shock:**
 1. **DO NOT TOUCH THEM!** You will get shocked too.

2. **TURN OFF THE POWER!** (Hit the main breaker or inverter switch).
 3. If you can't, use an **insulator** (a dry piece of wood, a dry cloth) to *push* the victim off the live wire.
 4. Call for help and administer CPR if trained.
- **If you get battery acid on your skin/eyes:**
 1. **FLUSH WITH WATER!** Run to a tap and flush the area with cool, running water for at least 15 minutes.

Evaluation:

1. List the three main types of hazards in solar installation.
2. What does PPE stand for?
3. What 3 pieces of PPE are most important for solar? (Goggles, Gloves, Boots).
4. Why must you *never* smoke or create sparks near a battery?
5. What is the *first* thing you should do if your co-worker is getting an electric shock?

Assignment:

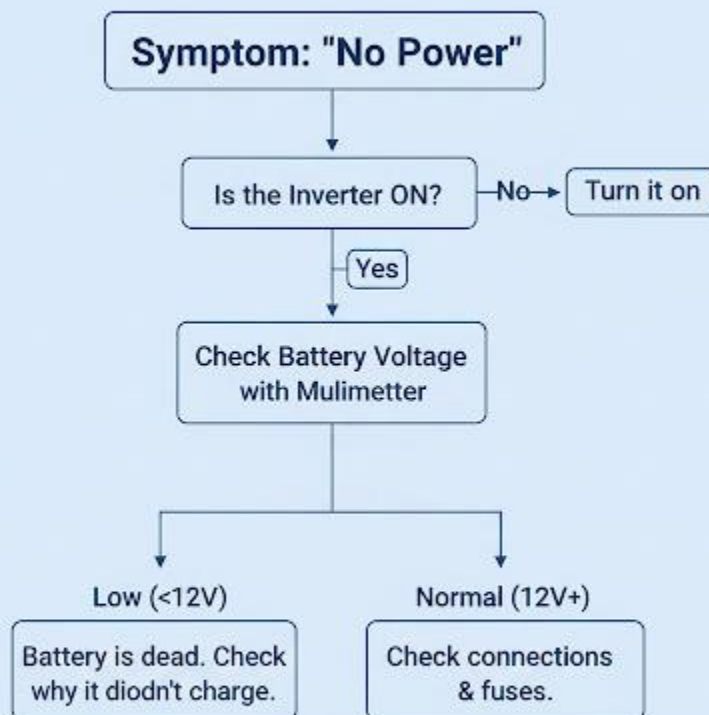
1. What is a "DC arc flash," and why is it so dangerous?
2. Why is it safer to work on live circuits with only one hand?
3. Describe the safe procedure for handling batteries (mention ventilation and tools).
4. What is the first aid response for getting battery acid in your eyes?
5. Why should you cover a solar panel with a blanket before you wire it?

WEEK 10: SOLAR PV MAINTENANCE AND TROUBLESHOOTING

Topic: Solar PV Maintenance and Troubleshooting

Breakdown (Subtopics):

- Meaning of maintenance
- Types: Preventive and corrective maintenance
- Common solar PV faults and causes
- Steps in troubleshooting a solar system
- Practical: Maintenance check on a sample setup



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Maintenance."
2. Differentiate between preventive and corrective maintenance.
3. List three common faults in a solar system and their likely causes.
4. Describe the logical steps of troubleshooting using a multimeter.
5. Perform a simple maintenance check.

Lesson Content:

1. Meaning of Maintenance

- **Definition:** Maintenance involves performing checks, cleaning, and repairs on a system to ensure it continues to operate **safely** and **efficiently**.
- **Importance:** A solar system is a long-term investment. Just like a car, it needs regular service to last 20+ years. Good maintenance *prevents* faults from happening.

2. Types of Maintenance

- **A. Preventive Maintenance (PM):**
 - **Definition:** This is the *scheduled* maintenance you do *before* a problem occurs. This is the main job of a maintenance technician.
 - **PM Checklist:**
 - **Panels:** Clean panel glass (Term 1, Week 8).
 - **Shading:** Check for new tree growth that is causing shadows.
 - **Wiring:** Check all connections (at panel, controller, battery) to ensure they are *tight*.
 - **Batteries (Lead-Acid):** Check battery terminals for *corrosion* (a white, powdery substance). Clean with a wire brush and water/baking soda.
 - **Batteries (Flooded):** Check the *water level* and top up with **distilled water** only.
- **B. Corrective Maintenance (CM):**

- **Definition:** This is the *unscheduled* maintenance you do *after* a problem has occurred. This is "troubleshooting" or "repair."
- *Example:* A fuse has blown, and you must find out why and replace it.

3. Common Solar PV Faults and Causes | Fault / Symptom | Likely Cause(s) | :--- | :--- | 1.

No power in the house. | • Inverter is OFF or tripped. • Battery is dead (over-discharged). • Main fuse/breaker is blown. | **2. Battery is not charging.** | • Panel wire is cut or disconnected. • Charge controller is damaged (often by reversed polarity). • Panel is completely covered in dust. • Fuse between panel and controller is blown. | **3. Low power / Short runtime.** | • Panels are dirty or shaded. • Battery is old and has lost its capacity. • You are using *more* power than the system was designed for. | **4. Inverter beeping.** | • **Overload:** You turned on too many appliances (e.g., fridge + electric iron). • **Low Battery:** The controller has cut off the power. |

4. Steps in Troubleshooting (Corrective Maintenance)

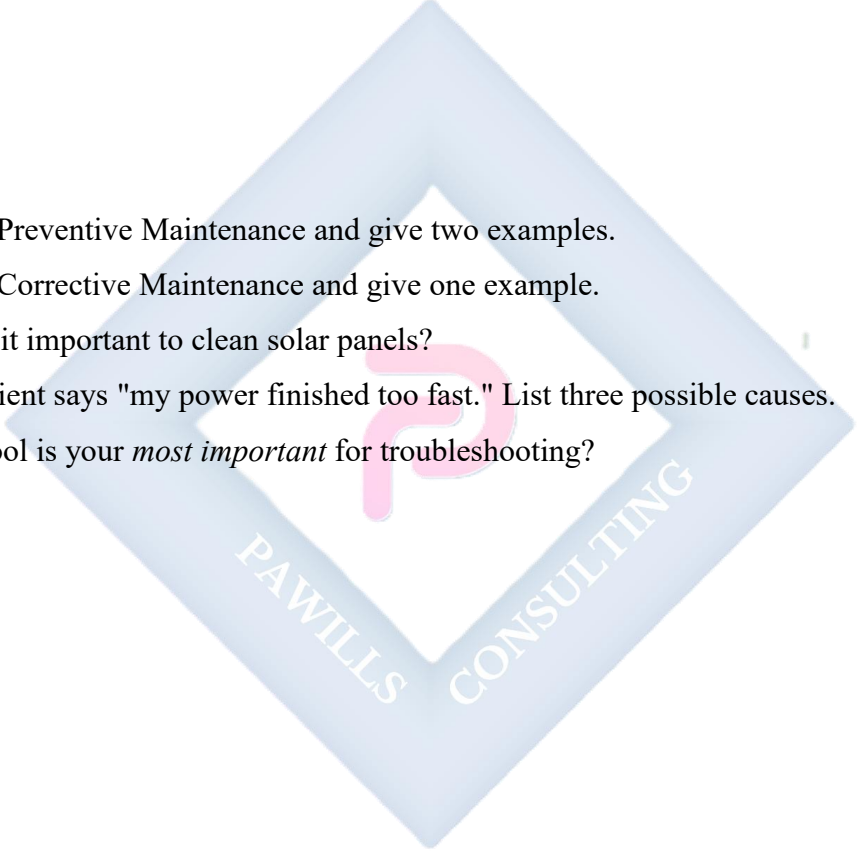
- **Rule:** Be logical. Do not guess. Use your **multimeter**.
- **Scenario:** The battery is not charging.
 - **Step 1 (The Panel):** Go to the controller. Disconnect the PV wires. Set your multimeter to **DC Voltage (V_—)**. Test the wires from the panel.
 - *Result?* Do you see the correct voltage (e.g., 18V)?
 - *If YES:* The panels are working. The problem is the controller or battery.
 - *If NO:* The problem is the panel (or the wire from the panel). Go to the roof.
 - **Step 2 (The Battery):** Go to the controller. Set multimeter to **DC Voltage (V_—)**. Test the BAT terminals.
 - *Result?* Does it read 12V?
 - *If YES:* The battery is connected.
 - *If NO:* The wire from the battery is broken, or the battery fuse is blown.
 - **Step 3 (The Controller):** If the panels are OK (Step 1) and the battery is OK (Step 2), but the battery is *still* not charging... then the **charge controller itself is faulty** and must be replaced.

Evaluation:

1. What is the goal of maintenance?
2. What is the difference between preventive and corrective maintenance?
3. List two *preventive* maintenance tasks.
4. Your client says their battery isn't charging. What is the *first* component you should test with your multimeter?
5. An inverter is beeping loudly. What are the two most likely reasons?

Assignment:

1. Define Preventive Maintenance and give two examples.
2. Define Corrective Maintenance and give one example.
3. Why is it important to clean solar panels?
4. Your client says "my power finished too fast." List three possible causes.
5. What tool is your *most important* for troubleshooting?

A large, light blue diamond-shaped watermark logo is centered on the page. Inside the diamond is a pink stylized 'P' logo. The text 'PAWILLS' is written in white along the bottom-left edge of the diamond, and 'CONSULTING' is written in white along the bottom-right edge.

PAWILLS CONSULTING

WEEK 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Review of all topics covered during the term

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall all major components and their functions.
2. Explain the processes of Load Estimation and System Sizing.
3. Review the key installation and safety procedures.
4. Discuss the common faults and maintenance procedures.

Lesson Content:

1. Summary of Term 2 Topics (The "Design & Install" Term):

- **Week 1 (Review):** We recapped Term 1 (Components, AC/DC, Series/Parallel, Safety).
- **Week 2 (Batteries):** The "storage tank."
 - Types: Flooded (cheap, needs water), Gel/AGM (sealed), **Lithium** (best, 80-100% DoD, long life).
 - **DoD (Depth of Discharge):** Lead-Acid is 50%, Lithium is 80%+.
- **Week 3 (Inverters):** The "converter" (DC to AC).
 - **MSW (Modified):** Cheaper, "dirty" power. Bad for fans/TVs.
 - **PSW (Pure Sine):** High quality, "clean" power. Safe for all appliances.
- **Week 4 (Load Estimation):** The "math."
 - **Power (W):** How *fast* energy is used.
 - **Energy (Wh):** How *much* energy is used. (Energy = Power × Time).
 - A **Load Audit** (listing all appliances) is the *first step* of any design.
- **Week 6 (System Sizing):** The "design."
 - **Inverter Size (W):** Based on *Running Watts*.
 - **Battery Size (Ah):** Based on *Energy (Wh)* and *Autonomy*.

- **Panel Size (Wp):** Based on *Energy (Wh)* and *Peak Sun Hours*.
- **Week 8 (Installation):**
 - **Site Survey:** Check for **South-facing** roof and **Shading**.
 - **Connection Order:** 1. Battery, 2. Panel.
 - **Commissioning:** Testing the system to make sure it works.
- **Week 9 (Safety):**
 - **Hazards:** Electric Shock (DC Arc), Battery (Acid, Gas), Falling.
 - **PPE:** Goggles, Insulated Gloves, Rubber-soled Boots.
 - **Golden Rules:** Use insulated tools, cover panels, *never* cause a spark near a battery, use fuses.
- **Week 10 (Maintenance):**
 - **Preventive (Before):** Cleaning panels, checking water, tightening terminals.
 - **Corrective (After):** Troubleshooting with a **multimeter**.
 - **Common Faults:** Dirty panels, loose connections, dead battery, tripped breaker.

2. Class Activity: Q&A Review Session (Teacher asks rapid-fire questions to test recall of all key concepts).

- "What is the best battery type?" (Lithium)
- "What is the best inverter type?" (Pure Sine Wave)
- "What is the first step of designing a system?" (Load Audit)
- "What unit do we use for Energy?" (Watt-hours)
- "What unit do we use for Power?" (Watts)
- "What tool do you use to find a fault?" (Multimeter)
- "What is the biggest cause of power loss?" (Dirty panels / Shading)

Evaluation: This week's evaluation is the Second Term Examination.

Assignment:

1. Study all notes from Week 1 to Week 10 for the examination.
2. What is the difference between a Pure Sine Wave and a Modified Sine Wave inverter?

3. What is the difference between a Lead-Acid battery and a Lithium-ion battery (mention DoD and Lifespan)?
4. A 100W TV runs for 5 hours. How much Energy (in Wh) does it use?
5. Based on Q4, if you use a 12V Lead-Acid battery (50% DoD), what is the *minimum* battery size (in Ah) you need? (Hint: Use the Sizing formula).
 - (Answer: $(100W * 5h) / (0.50 \text{ DoD} * 12V) = 500 / 6 = 83.3 \text{ Ah}$)



**THIRD TERM - JSS 2 SOLAR PHOTOVOLTAIC (PV) INSTALLATION AND
MAINTENANCE LESSON NOTES**



WEEK 1: REVIEW OF SECOND TERM WORK

Topic: Review of Second Term Work

Breakdown (Subtopics):

- Recap of installation and maintenance concepts
- Review of inverter, battery, and system sizing
- Identifying solar system components
- Class discussion and short quiz

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall the three main types of solar batteries and their key characteristics (DoD, Lifespan).
2. Differentiate between a Pure Sine Wave and a Modified Sine Wave inverter and state which appliances they can power.
3. Define "Load Audit" and "System Sizing" and explain why they are critical.
4. List the key steps in a safe installation procedure and a preventive maintenance check.
5. Verbally demonstrate understanding of these concepts in a class discussion.

Lesson Content:

1. Recap of Key Components (Inverter & Battery)

- **Batteries (The "Storage Tank"):**
 - **Flooded Lead-Acid:** Cheapest, but requires maintenance (topping with distilled water) and has a low lifespan (~500 cycles) and low DoD (~50%).
 - **Gel/AGM (Sealed):** Maintenance-free, but more expensive and can be damaged by heat. DoD is ~50%.
 - **Lithium-ion (LiFePO4):** The best. Very long lifespan (4000+ cycles), high DoD (80-100%), lightweight, and maintenance-free, but has the highest initial cost.
- **Inverters (The "Converter" - DC to AC):**

- **Modified Sine Wave (MSW):** Cheaper, "blocky" power. **Dangerous** for sensitive electronics (TVs, fridges, fans). Only use for simple bulbs or heaters.
- **Pure Sine Wave (PSW):** High quality, "clean" power identical to NEPA. **Safe for all appliances.** This is the professional standard.

2. Review of System Sizing (The "Math") This is the most critical skill for an installer.

1. Step 1: Load Estimation (Load Audit):

- First, find the **Total Daily Energy (Wh)** needed.
- Formula: **Energy (Wh) = Power (W) × Time (h).**
- You must do this for *every* appliance (bulb, fan, TV) and *add* them all together.

2. Step 2: Sizing the Components:

- **Inverter Size (W):** Based on the total *Running Watts* (all appliances on at once) + 25% safety margin. (e.g., 265W load needs a 331W inverter, so you buy a 500W).
- **Battery Bank Size (Ah):** Based on *Total Daily Energy (Wh)*.
 - Formula: $(\text{Energy (Wh)} \times \text{Days of Autonomy}) / (\text{DoD} \times \text{System Voltage})$
 - (e.g., $1460\text{Wh} / (0.50 \text{ DoD} \times 12\text{V}) = 243.3 \text{ Ah}$).
- **Solar Panel Array Size (Wp):** Based on *Total Daily Energy (Wh)*.
 - Formula: $\text{Total Energy (Wh)} / (\text{Peak Sun Hours} \times \text{System Efficiency})$
 - (e.g., $1460\text{Wh} / (4.5 \text{ PSH} \times 0.77) = 421 \text{ W}$).

3. Review of Installation & Maintenance

• Installation:

- **Site Survey:** Find a **South-facing** roof with **no shading**.
- **Safety:** Use PPE (goggles, gloves, boots). Never smoke near batteries (explosive gas).
- **Wiring Order:** **CONNECT BATTERY FIRST** to the charge controller.
- **Troubleshooting:** The **Multimeter** is your best tool. The most common faults are **Reversed Polarity (+ to -)**, **Loose Connections** (fire hazard), and **Dirty/Shaded Panels**.

• Maintenance:

- **Preventive:** The scheduled work you do to stop problems.
 - **Task 1: Cleaning panels** to remove dust.
 - **Task 2: Tightening** all wire connections.
 - **Task 3:** Checking battery water/terminals for corrosion.
- **Corrective:** The repair work done *after* a fault.

4. Class Discussion (Assessment)

- "What is the difference between Power (Watts) and Energy (Watt-hours)?"
- "Why is a Lithium battery (200Ah) better than a Lead-Acid battery (200Ah)?" (Usable capacity/DoD).
- "Why can't you use a cheap inverter for a fridge?" (MSW will damage the motor).
- "What are the three main things you look for in a site survey?" (South-facing, no shade, strong roof).
- "What is the first component you disconnect from a controller?" (The panel).

Evaluation: (This is an oral evaluation based on the class discussion)

1. Explain the difference between a PSW and MSW inverter.
2. Explain the difference between a Lithium and Lead-Acid battery, mentioning DoD.
3. What are the three sizing formulas for (a) Inverter, (b) Battery, and (c) Panels?
4. What are the two main types of maintenance?
5. What are three major safety hazards during installation?

Assignment:

1. Define "Load Audit" and "System Sizing."
2. A 100W fan runs for 8 hours. How much Energy (in Wh) does it consume?
3. Using the 800Wh from Q2, calculate the *minimum* battery bank (in Ah) needed for a 12V system with 50% DoD and 1 day of autonomy.
4. List the correct order for *connecting* a solar system (Panel, Battery, Inverter, Controller).
5. List three *preventive* maintenance tasks.

WEEK 2: ENERGY STORAGE SYSTEMS

Topic: Energy Storage Systems

Breakdown (Subtopics):

- Meaning and importance of energy storage
- Types of energy storage systems: chemical, mechanical, thermal
- Batteries as energy storage devices in PV systems
- Factors affecting energy storage efficiency
- Practical: Observing battery charge/discharge patterns



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Energy Storage."
2. State why energy storage is critical for renewable energy.
3. List and differentiate between chemical, mechanical, and thermal storage.
4. Identify batteries as the primary form of chemical storage for PV.
5. Define "round-trip efficiency" and explain how it affects a system.

Lesson Content:

1. Meaning and Importance of Energy Storage

- **Definition:** An **Energy Storage System (ESS)** is any device or technology that can capture energy, store it for a period of time, and release it on demand.
- **Importance:** Renewable energy sources like solar and wind are *intermittent* (they are not available 24/7).
 - Solar panels work only during the day.
 - Wind turbines work only when the wind blows.
- The primary challenge of renewable energy is *not* production, it is *storage*. We need to "save" the energy produced at 12:00 noon for use at 8:00 p.m. Energy storage (like a battery) solves this intermittency problem and makes clean energy reliable.

2. Types of Energy Storage Systems There are many ways to store energy, mostly for large-scale grid use:

- **A. Chemical Storage (The most common for PV):**
 - **How it works:** Stores energy in chemical bonds. A reversible chemical reaction releases electrons (electricity).
 - **Example: Batteries.** (Lead-Acid, Lithium-ion). This is the only type that is practical for a home solar (PV) system.
- **B. Mechanical Storage:**
 - **How it works:** Stores energy in the *motion* or *position* of an object.

- **Example 1: Pumped Hydro Storage.** (This is the world's largest form of grid storage). Two reservoirs (lakes) at different heights. When there is excess solar power, it is used to *pump* water from the low lake to the high lake. At night, the water is released, flowing *down* through a hydro-turbine (like Kainji dam) to generate electricity.
- **Example 2: Flywheels.** A heavy wheel is spun at super-high speed (using electric motors). The spinning wheel stores kinetic energy. To get power back, the wheel's motion is used to turn a generator.
- **C. Thermal Storage:**
 - **How it works:** Stores energy as *heat* or *cold*.
 - **Example 1:** A simple **hot water tank** (like in a solar thermal system) is a thermal storage device.
 - **Example 2 (Grid-scale):** Molten Salt. Large solar thermal farms use mirrors to focus sunlight on a tower, heating salt to over 500°C. This molten salt stays hot all night, and its heat is used to boil water, create steam, and turn a turbine 24/7.

3. Batteries: The Best Fit for PV Systems While mechanical and thermal storage are excellent for the grid, they are not practical for a house. **Batteries (Chemical Storage)** are the standard for PV systems because they are:

- **Scalable:** You can use one small battery or one hundred large ones.
- **Responsive:** They provide power instantly.
- **Compact:** They fit inside a home.

4. Factors Affecting Energy Storage Efficiency

- **Round-Trip Efficiency:**
 - **Definition:** This measures how much energy you get *back* compared to how much you put *in*. Some energy is always lost as heat during charging and discharging.
 - **Example:** You put 100 Watts into a battery, but you only get 85 Watts back out. The Round-Trip Efficiency is 85%.

- **Lead-Acid:** ~70-85% efficient. (If you put in 100Wh, you get 70-85Wh back).
- **Lithium-ion:** ~95-99% efficient. (If you put in 100Wh, you get 95-99Wh back).

This is a major advantage.

- **Temperature:** Batteries hate extreme heat. High temperatures (common in Nigeria) will *significantly* reduce the lifespan (number of cycles) of all batteries, especially Lead-Acid. Batteries should always be kept in the coolest, most ventilated place possible.
- **Depth of Discharge (DoD):** (Review) Regularly discharging a battery below its recommended DoD (e.g., 50% for lead-acid) will rapidly destroy its efficiency and lifespan.

5. Practical: Observing Charge/Discharge

- (Teacher connects a battery to a multimeter). "The voltage is 12.8V. This battery is full."
- (Teacher connects a DC fan (a load) to the battery). "Watch the multimeter. The voltage is now dropping to 12.5V... 12.4V. The battery is *discharging*."
- (Teacher connects a solar panel to the battery via a controller). "Now, with the panel in the sun and the fan off, watch the voltage. It is rising... 12.9V... 13.0V. The battery is *charging*."

Evaluation:

1. Why is energy storage so important for renewable energy sources like solar?
2. What are the three main types of energy storage?
3. What is the most common type of storage used for home PV systems?
4. What is "round-trip efficiency"?
5. Why is a Lithium-ion battery more efficient than a Lead-Acid battery?

Assignment:

1. Define "Energy Storage System" and "Intermittency."
2. Explain "Pumped Hydro Storage" in your own words.
3. Explain "Thermal Storage" using the molten salt example.
4. What is "round-trip efficiency"? Why is 95% better than 80%?

5. How does high temperature (like in Nigeria) affect a solar battery?

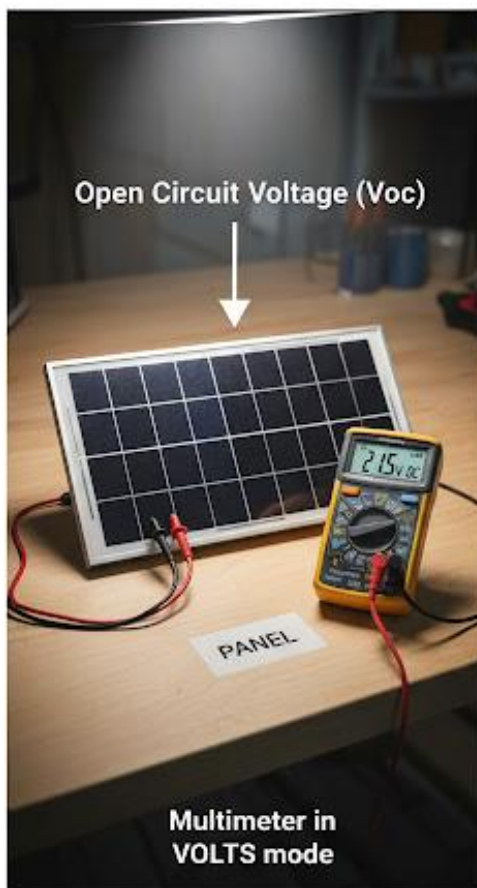


WEEK 3: SOLAR SYSTEM PERFORMANCE EVALUATION

Topic: Solar System Performance Evaluation

Breakdown (Subtopics):

- Meaning and importance of performance evaluation
- Parameters to monitor (voltage, current, efficiency)
- Tools and instruments for system testing
- Recording and interpreting data
- Practical: Measuring voltage and current from a panel



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Performance Evaluation" and state its importance.
2. List the three key electrical parameters used for evaluation (V, I, P).
3. Define "Solar Irradiance" and the tool used to measure it.
4. Define Panel "Open Circuit Voltage (Voc)" and "Short Circuit Current (Isc)."
5. Correctly use a multimeter to measure Voc and Isc from a small panel.

Lesson Content:

1. Meaning and Importance of Performance Evaluation

- **Definition:** Performance evaluation is the process of systematically **measuring and analyzing** a solar PV system to see if it is working correctly and producing the amount of power it was *designed* to produce.
- **Importance:**
 - **For the Client:** To confirm they got what they paid for.
 - **For the Installer:** To prove the installation was done correctly, and to find faults (troubleshooting). A system may be "working" (powering a bulb) but "underperforming" (only producing 50% of its expected power).

2. Parameters to Monitor

1. **Voltage (V):** Is the electrical "pressure" correct?
2. **Current (A):** Is the "flow" of electricity strong?
3. **Power (W):** The *instant* output. **Power (W) = Voltage (V) × Current (A).**
4. **Energy (J):** The *total* output over time (e.g., in one day).
5. **Solar Irradiance:** The amount of *sunlight power* hitting the panel. You can't blame a panel for low output if it's a cloudy day.

3. Tools and Instruments for System Testing

- **Multimeter:** (Review) Used to measure Voltage (V) and Current (A).

- **Clamp Meter:** A special type of multimeter with a "clamp" that can measure high currents (especially AC) *without* disconnecting the wire.
- **Pyranometer:** A scientific tool used to measure **Solar Irradiance** (sunlight intensity), measured in **W/m²** (Watts per square meter). At noon on a clear day, irradiance is about 1000 W/m².
- **Monitoring App:** Most modern hybrid inverters connect to WiFi and have an app that shows you all the data (panel power, battery level, grid use) in real-time on your phone.

4. Recording and Interpreting Data (Key Panel Specs) Every solar panel has a "datasheet" with its factory-tested values. To test a panel, you compare its *real-world* measurements to its *datasheet* values. The two most important values to test are:

- **Open Circuit Voltage (Voc):**
 - **Definition:** The *maximum voltage* a panel can produce when it is in the sun but **not connected** to anything (an "open circuit").
 - **Use:** This is the first test you do. It proves the panel is "alive."
- **Short Circuit Current (Isc):**
 - **Definition:** The *maximum current (Amps)* a panel can produce. You measure this by *briefly* "shorting" the positive and negative wires together through an ammeter.
 - **Use:** This tests if the panel's cells are all working and producing full power.

5. Practical: Measuring Voc and Isc (Teacher takes a small 12V panel into the sun. **Safety goggles must be worn.**)

- **A. Measuring Open Circuit Voltage (Voc):**
 1. Set the multimeter to **DC Voltage (V=)**.
 2. Touch the Red probe to the panel's (+) wire.
 3. Touch the Black probe to the panel's (-) wire.
 4. **Result:** A 12V panel will show a Voc of **~18-22V**. This is normal. (This high voltage is why you *must* use a charge controller).
- **B. Measuring Short Circuit Current (Isc):**
 1. Set the multimeter to the **DC Amps (A=)** setting (e.g., 10A).

2. (This is the "short circuit").
 3. Touch the Red probe to the panel's (+) wire.
 4. Touch the Black probe to the panel's (-) wire.
 5. **Result:** The meter will show the max current, e.g., **5.5A**.
 6. *Immediately* disconnect. Do not hold this for more than a few seconds.
- **Conclusion:** If the measured Voc and Isc match the panel's datasheet, the panel is healthy. If they are very low, the panel is faulty or dirty.

Evaluation:

1. What is "performance evaluation"?
2. What is the formula for calculating Power? ($W = V \times A$).
3. What is "Open Circuit Voltage (Voc)"?
4. What is "Short Circuit Current (Isc)"?
5. What tool is used to measure solar irradiance (sunlight intensity)?

Assignment:

1. Why is it important to evaluate a solar system's performance?
2. What is the difference between a multimeter and a clamp meter?
3. A 12V panel has a Voc of 20V and an Isc of 8A. What is its *maximum theoretical power* (W)? (Hint: $P = V \times I$).
4. Describe, step-by-step, how to measure the **Voc** of a solar panel.
5. You test a 12V panel and its Voc is only 5V. What are two possible reasons for this low reading? (e.g., "It is damaged," "It is in the shade," "It is very dirty").

WEEK 4: FAULT DETECTION IN PV SYSTEMS

Topic: Fault Detection in PV Systems

Breakdown (Subtopics):

- Common PV system faults (panel shading, loose connection, battery failure)
- Troubleshooting techniques
- Step-by-step guide to fault tracing
- Documentation and reporting of faults
- Demonstration: Identifying faults in a sample solar circuit



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Troubleshooting."
2. List three of the most common PV system faults.
3. Explain the negative impact of "shading" on a system.
4. Describe a logical, step-by-step process for tracing a fault.
5. Define what a "fault report" is and why it is important.

Lesson Content:

1. What is Troubleshooting (Fault Detection)?

- **Definition:** Troubleshooting is the logical, systematic search for the *cause* of a problem in order to *solve* it. It is "corrective maintenance."
- **The Rule: DO NOT GUESS.** Start from the symptom (e.t., "no power") and logically test each component one by one.

2. Common PV System Faults

- **A. Panel Shading (The "Silent Killer"):**
 - **Fault:** A single tree branch, a TV antenna, or even bird droppings casting a shadow over even *one small part* of a single panel.
 - **Effect:** In a series-wired string of panels, if one panel is shaded, the output of the *entire string* can drop to almost zero. It is a major performance killer.
- **B. Loose Connections (The "Fire Starter"):**
 - **Fault:** A terminal screw on the battery, inverter, or controller that is not fully tightened.
 - **Effect:** The loose connection creates high **resistance**. As high current (Amps) tries to pass, this resistance generates extreme **heat**, melting the wire, melting the terminal, and potentially **starting a fire**.
- **C. Battery Failure (The "End of Life"):**
 - **Fault:** The battery has reached the end of its cycle life, or it has been damaged by over-discharging (below 50% DoD) or overheating.

- **Effect:** The battery will *appear* to charge fully (e.g., 13V) but will "die" (voltage drops) as soon as you put a small load on it. It can no longer *hold* a charge.
- **D. Reversed Polarity:** (Review) Connecting (+) to (-). This instantly *destroys* the controller or inverter.
- **E. Blown Fuses / Tripped Breakers:** This is a *symptom*, not the cause. The fuse blew for a reason (e.g., a short circuit or an overload).

3. Step-by-Step Guide to Fault Tracing

- **Symptom:** The client calls and says, "My solar is not working at night."
- **Logical Steps (Troubleshooting):**
 1. **Question the User:** "Is the inverter on? Is it beeping? Did you add a new appliance (like a freezer)?" (The user's answers will guide you).
 2. **Inspect (Look & Listen):** Go to the system. Are there any error codes on the inverter? Can you smell melting plastic (a loose connection)? Are any breakers tripped?
 3. **Check the Battery (The "Heart"):**
 - Test the battery voltage with a multimeter. Is it 0V? Is it 10.5V? (This means it's dead, and the *real* problem is why it's dead).
 4. **Check the Panel (The "Generator"):**
 - Is it daytime? Go to the controller. Is the "PV" or "Charge" light on?
 - If No: Use your multimeter to test the panel wires (Voc). Is there voltage?
 - *If No:* The fault is on the roof (panel is shaded, wire is cut, panel is dead).
 5. **Check the Controller (The "Brain"):**
 - Is the panel giving power (e.t., 20V) and the battery is low (e.g., 12V), but the controller is not passing the charge?
 - *Conclusion:* The controller is faulty.

4. Documentation and Reporting of Faults

- **Definition:** After you find and fix the problem, you must write a simple **Fault Report** for the client (and for your own records).

- **Importance:**

- **Professionalism:** It shows the client what you did.
- **Records:** If the same problem happens again, you have a record.

- **Report should state:**

1. **Date:**
2. **Symptom Reported:** (e.g., "No power at night").
3. **Fault Found:** (e.g., "Loose negative connection on battery terminal was overheating").
4. **Action Taken:** (e.g., "Disconnected system, cleaned terminal, and re-tightened all connections").
5. **Result:** (e.g., "System is now charging and operating normally").

5. Demonstration:

- (Teacher sets up a small solar circuit: Panel -> Controller -> Battery -> Load).
- (Teacher creates a *fault*, e.g., disconnects the panel wire, or uses a "blown" fuse).
- (Students are asked to use a multimeter, following the logical steps, to find *where* the electricity stops).

Evaluation:

1. What is "troubleshooting"?
2. What is the *effect* of a small shadow on a large solar array?
3. Why is a loose connection so dangerous?
4. Your client says their battery isn't charging. What are the two *most likely* causes?
5. What is a "fault report," and why is it important?

Assignment:

1. Define Troubleshooting.
2. List three common PV system faults.

3. You are troubleshooting a "No Power" fault. You test the battery with a multimeter, and it reads 10.5V. What does this tell you? (The battery is dead, and the controller's over-discharge protection is active).
4. What is the *real* problem you must now find? (Why is the battery not getting charged?).
5. Write a sample (short) Fault Report for the following: The client's panels were dirty.
 - (Symptom: Low power. Fault Found: Panels covered in dust. Action: Cleaned panels. Result: System output increased by 25%).



WEEK 6: ENTREPRENEURSHIP IN SOLAR ENERGY

Topic: Entrepreneurship in Solar Energy

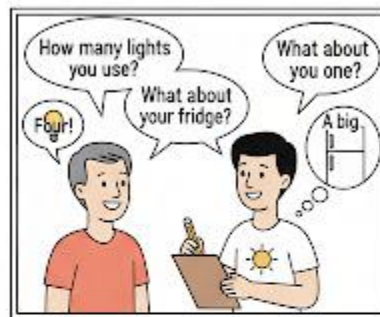
Breakdown (Subtopics):

- Meaning of entrepreneurship
- Solar business opportunities (installation, sales, maintenance)
- Customer service and communication
- Skills needed for successful solar entrepreneurship
- Role play: Acting as a solar consultant

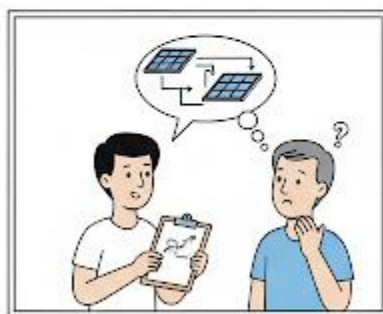
YOUR SOLAR CONSULTANT: ASKING THE RIGHT QUESTIONS!



Client



Solar Consultant
Student



Analyzing your energy needs



You can save 60%!

Custom Solutions for Maximum Savings!

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Entrepreneurship" (recap).
2. List and describe three business opportunities in the solar value chain.
3. Define "Customer Service" and list three qualities of a professional.
4. Differentiate between technical skills and "soft skills."
5. Participate in a role-play as either a client or a consultant.

Lesson Content: (This is a review and expansion of Term 1, Week 10)

1. Meaning of Entrepreneurship

- **Definition:** An entrepreneur is a person who identifies a **problem** (or need) in society, creates a business **solution**, and takes a financial **risk** to deliver that solution, hoping to make a **profit**.
- **Problem:** Unreliable, noisy, and expensive electricity from the grid and generators.
- **Solution:** Providing clean, reliable, and silent solar power.

2. Solar Business Opportunities (The "Value Chain")

1. Installation Technician (The Installer):

- This is the *skilled labor*. The person who performs the site survey, system sizing, and physical installation. This is the focus of this course.

2. Sales and Distribution (The Seller/Vendor):

- This person *imports, stocks, and sells* the components (panels, batteries, inverters) in a shop or online. They must have good business and marketing skills.

3. Maintenance Services (The "Doctor"):

- This entrepreneur offers "Solar Maintenance" contracts. They visit clients (e.g., 4 times a year) to perform *preventive maintenance* (cleaning panels, checking batteries).
- This is an excellent business model as it provides *recurring revenue* (steady income).

4. Consulting and System Sizing (The "Architect"):

- This is an expert who does *not* sell products. They are hired by a client to perform a professional load audit and design the *perfect* system. The client then takes this design to a *seller* to buy the exact products. This avoids being cheated by a seller.

3. Customer Service and Communication (Soft Skills)

- **Technical Skills:** Knowing *how* to wire a panel.
- **Soft Skills:** Knowing *how* to talk to the client.
- **In business, soft skills are just as important as technical skills.**
- **Key Soft Skills / Professionalism:**
 1. **Honesty and Transparency:** (Most important!) Never lie about a product's capability. Don't sell a 200Ah battery as a 300Ah. Don't sell MSW inverters as PSW. A good reputation is your best advertisement.
 2. **Communication:** You must be able to *listen* to the client's needs and *explain* complex ideas (like "DoD" or "sizing") in simple terms.
 3. **Punctuality:** Arrive when you say you will arrive.
 4. **Neatness:** A professional installation is *neat*. Wires are hidden in conduit (pipes), not dangling. A neat job *looks* safe and builds trust.
 5. **After-Sales Service:** Answer your phone when the client calls with a problem *after* you have been paid. This builds loyalty and referrals.

4. Role Play: Solar Consultant

- **Scenario:** One student is the "Client" (e.g., a small shop owner). One student is the "Solar Consultant" (Entrepreneur).
- **Consultant's Job:** Do not try to sell anything. Your job is to *ask questions* to perform a load audit.
 - "Good morning, sir. To design a system for you, I need to know what you want to power."
 - "How many lights do you have? What type? How many hours do you use them?"
 - "What about fans? How many? How many hours?"
 - "Do you have a fridge? Is it a freezer or a small bar fridge?"

- **Client's Job:** Answer the questions (e.g., "I have 4 bulbs, 1 fan, and 1 TV. I want power from 6 p.m. to 11 p.m.").

Evaluation:

1. What is a solar "entrepreneur"?
2. List three different business opportunities in the solar sector.
3. What is the difference between "technical skills" and "soft skills"?
4. What is the *most important* quality of a professional installer? (Honesty).
5. Why is a "neat" installation important?

Assignment:

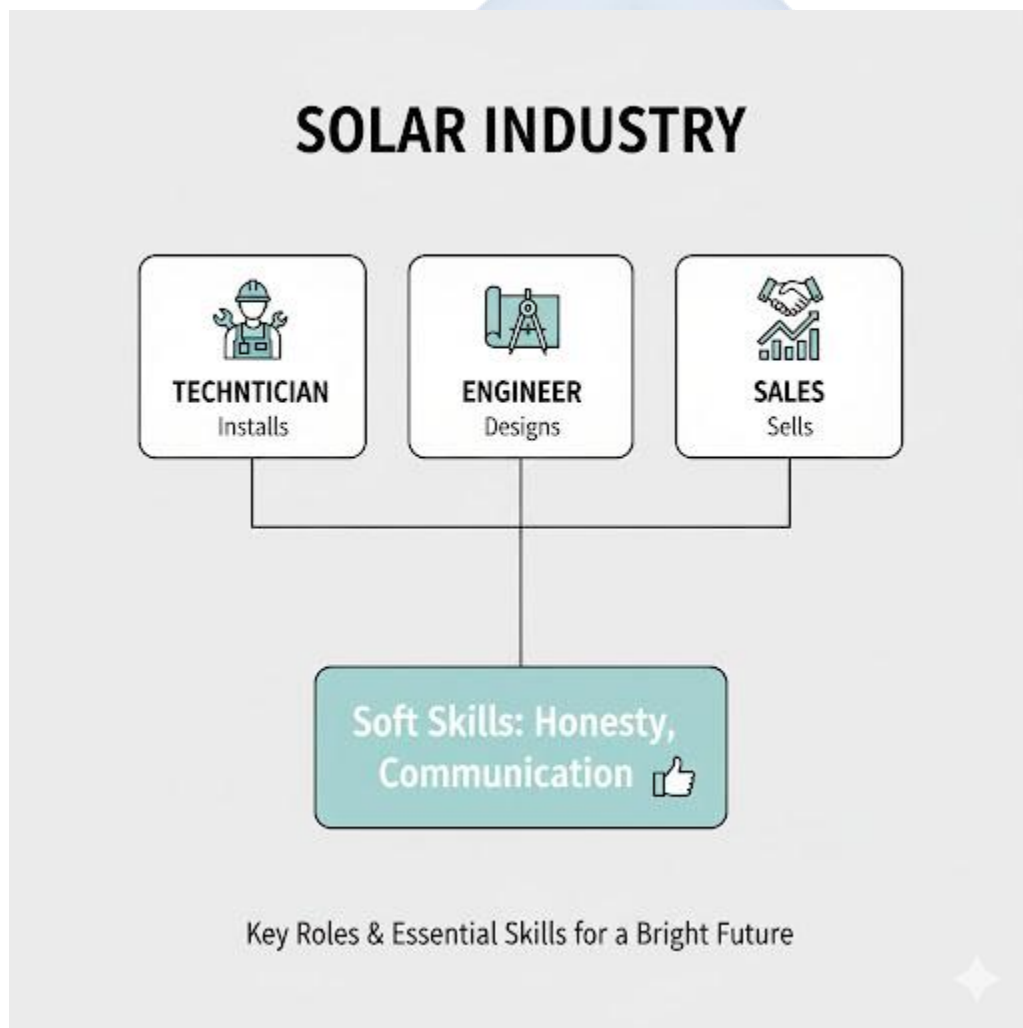
1. Define "Entrepreneurship."
2. List two "soft skills" and explain why they are important for a solar installer.
3. Imagine you are starting a solar installation business. Create a business name for it.
4. Who would be your "target customer"? (e.g., "Homes," "Small businesses," "Schools").
5. Write a short (3-4 sentences) "sales pitch" explaining to a potential customer *why* they should choose you (e.g., "I am honest, I use only Pure Sine Wave inverters, and my installations are neat and safe...").

WEEK 8: CAREER OPPORTUNITIES IN RENEWABLE ENERGY

Topic: Career Opportunities in Renewable Energy

Breakdown (Subtopics):

- Careers in solar and renewable energy industry
- Qualifications and skills required
- Institutions offering solar training
- Gender inclusivity and innovation in renewable energy
- Class presentation: “My dream solar career”



Performance Objectives: By the end of this lesson, students should be able to:

1. List at least five different career paths in the renewable energy sector.
2. Differentiate between a technician, an engineer, and a sales representative.
3. Identify the key skills (STEM) required for these careers.
4. Discuss the importance of gender inclusivity in the field.

Lesson Content:

1. Careers in the Renewable Energy Industry (This expands on Week 6. It's not just about starting your own business, but also about *working* for a large company).

- **1. Installation Technician:** (The "Hands-on" role) The skilled professional who physically installs and maintains PV systems.
- **2. Solar (PV) Engineer:** (The "Designer" role) An electrical engineer who designs large, complex systems (e.g., for factories or solar farms). They do the advanced calculations and drawings.
- **3. Sales Representative / Marketing Manager:** (The "People" role) The person who works for a solar company to find clients, explain the benefits, and sell systems. Requires good communication (soft skills).
- **4. R&D Scientist (Research & Development):** (The "Innovator" role) A scientist (e.g., with a Ph.D. in Chemistry or Physics) who works in a lab to invent the *next* generation of solar panels (e.g., more efficient, cheaper).
- **5. Data Analyst:** (The "Data" role) This person works for a large solar company (e.g., a solar farm) and analyzes the performance data from thousands of panels to spot trends and identify faults.
- **6. Renewable Energy Policy Advisor:** (The "Government" role) This person works for the government (e.g., Ministry of Energy) to create laws and policies that *encourage* the use of clean energy.

2. Qualifications and Skills Required

- **Foundation (Your Level):** A strong passion for **STEM** subjects:

- Science (especially Physics - for electricity)
- Technology (like this class)
- Engineering (understanding how things are built)
- Mathematics (for all the sizing and calculations)
- **Qualifications:**
 - **For a Technician:** Vocational school, trade certifications (like **NABCEP**, an international standard), and hands-on experience.
 - **For an Engineer/Scientist:** A University degree (B.Eng, M.Sc, Ph.D.) in Electrical Engineering, Physics, or Materials Science.
 - **For Sales/Policy:** A University degree in Business, Marketing, Law, or Economics.

3. Institutions Offering Solar Training

- Many organizations, private companies, and universities in Nigeria and around the world now offer specialized solar training courses, from short 2-week "bootcamps" to full engineering degrees. (e.g., The German GIZ program, various technical colleges).

4. Gender Inclusivity in Renewable Energy

- **The Past:** Technical fields like engineering and installation were often seen as "men's work."
- **The Present/Future:** This is changing fast. The solar industry needs the best brains, regardless of gender. Women are increasingly successful as solar engineers, installers, sales leaders, and technicians.
- **Importance:** Including everyone (men and women) brings more ideas, more innovation, and a larger workforce to solve our energy problems faster.

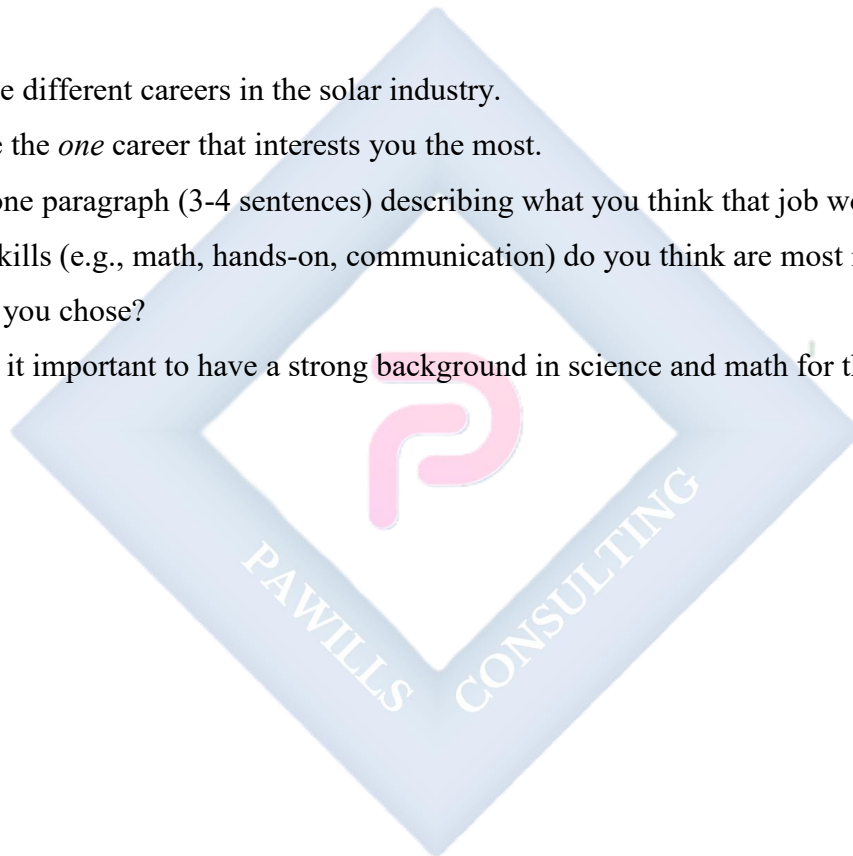
5. Class Presentation: "My Dream Solar Career" (Students are asked to briefly state which career (e.t., Technician, Engineer, Sales, Scientist) they find most interesting and why).

Evaluation:

1. List three different careers in the renewable energy industry.
2. What is the main difference between a "Technician" and an "Engineer"?
3. What does STEM stand for?
4. Why is "gender inclusivity" (including women) important for the solar industry?
5. What kind of qualifications would an R&D Scientist need?

Assignment:

1. List five different careers in the solar industry.
2. Choose the *one* career that interests you the most.
3. Write one paragraph (3-4 sentences) describing what you think that job would be like.
4. What skills (e.g., math, hands-on, communication) do you think are most important for the job you chose?
5. Why is it important to have a strong background in science and math for these careers?

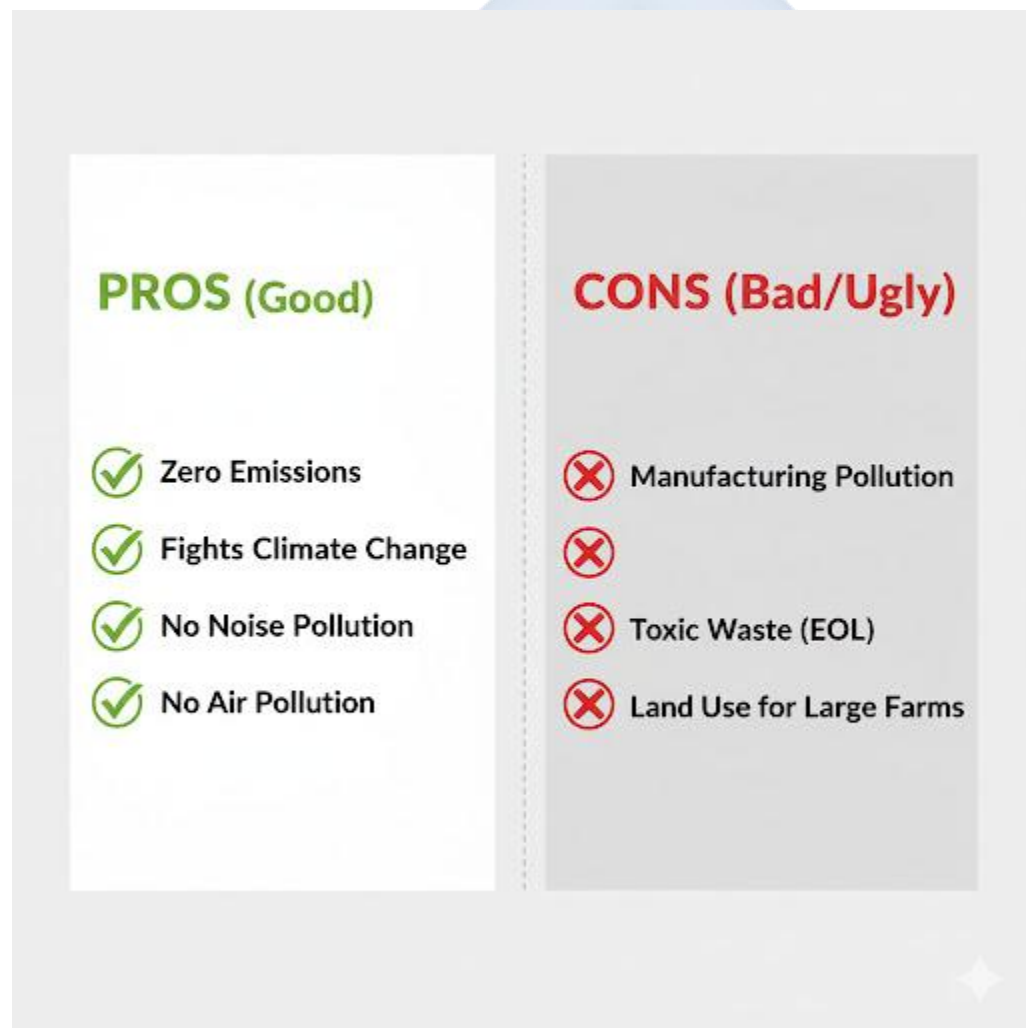


WEEK 9: ENVIRONMENTAL IMPACT OF SOLAR ENERGY

Topic: Environmental Impact of Solar Energy

Breakdown (Subtopics):

- Positive impacts: clean energy, reduced carbon footprint
- Negative impacts: waste disposal, land use
- Recycling of panels and batteries
- Sustainable solar practices
- Debate: “Solar Energy is 100% Environmentally Friendly”



Performance Objectives: By the end of this lesson, students should be able to:

1. List three major *positive* environmental impacts of solar energy.
2. List two *negative* environmental impacts of solar energy.
3. Define "carbon footprint."
4. Explain the importance of recycling solar panels and batteries.
5. Form an educated argument for or against the debate topic.

Lesson Content:

1. Positive Environmental Impacts (The "Good") This is the primary reason we use solar energy.

- **1. Reduces Carbon Footprint (Fights Climate Change):**

- **Carbon Footprint:** The total amount of greenhouse gases (like Carbon Dioxide, CO₂) released by a person, company, or country.
- Burning petrol in generators or gas in power plants (non-renewable) releases *tons* of CO₂, creating a large carbon footprint and causing global warming.
- **Solar's Impact:** A solar panel generates electricity with **zero CO₂ emissions**. It has a carbon footprint of *zero* during its 25+ years of operation.

- **2. Eliminates Air Pollution:**

- Generators also release soot (black smoke), sulphur dioxide, and nitrogen oxides. These cause *smog* and serious health problems (like asthma and lung cancer).
- Solar power is **100% clean**, producing no smoke or pollutants.

- **3. Eliminates Noise Pollution:**

- Generators are extremely loud and create a stressful environment.
- Solar systems (panels, inverters, batteries) are **completely silent**.

- **4. Water Conservation:**

- Traditional thermal (gas/coal) power plants use *billions* of litres of water for steam and cooling.
- Solar PV uses *no water* to generate electricity.

2. Negative Environmental Impacts (The "Bad" and "Ugly") Solar energy is *clean*, but it is not *perfect*. We must be honest about its challenges.

- **1. Manufacturing (The "Bad"):**

- Making solar panels is an industrial process that uses a lot of *energy* (which may come from dirty sources) and some *toxic chemicals* (like hydrofluoric acid).
- Mining for lithium and cobalt (for batteries) also has a significant environmental impact.

- **2. End-of-Life (EoL) Waste (The "Ugly"):**

- **The Problem:** What do we do with panels and batteries after their 20-year (panel) or 5-year (lead-acid) lifespan?
- **Solar Panels:** Contain small amounts of heavy metals like lead, cadmium, and chromium. If dumped in a landfill, these can leak into the groundwater and poison it.
- **Batteries: Lead-Acid** batteries are *extremely toxic* and must *never* be dumped. Lead is a neurotoxin.

- **3. Land Use:**

- A single home system is small. But a large *solar farm* (to power a city) can take up *hundreds* of acres, which could have been used for farming or as a natural habitat.

3. Recycling and Sustainable Solar Practices

- **The Solution:** We must create a "circular economy" for solar.
- **Battery Recycling:** Lead-Acid batteries are already highly recyclable (up to 98% of the lead can be recovered and reused). This is a major industry.
- **Panel Recycling:** This is a new and growing industry. Specialized plants can safely separate the glass, aluminum (frame), and the valuable silicon and silver from inside the panel, so they can be used to make *new* panels.
- **Sustainable Practice:** As an installer, you *must* have a plan to return dead batteries to a certified recycling center, not dump them.

4. Debate: "Solar Energy is 100% Environmentally Friendly"

- **FOR:** "Yes, it is. Its operational footprint is zero. It produces no CO₂, no smoke, and no noise. It is the best solution we have to fight climate change."
- **AGAINST:** "No, it is not 100% friendly. The manufacturing process causes pollution, and the waste (dead panels and batteries) is toxic. It is *cleaner* than generators, but it is not *perfect*."

Evaluation:

1. What is a "carbon footprint"?
2. List three positive environmental impacts of solar PV.
3. List two *negative* environmental impacts of solar PV.
4. Why is it dangerous to dump old lead-acid batteries in the ground?
5. What does it mean to "recycle" a solar panel?

Assignment:

1. Define "Carbon Footprint."
2. Explain why solar power is better for the environment than a petrol generator (mention CO₂ and noise).
3. What are the two main *negative* environmental impacts of solar PV?
4. Why is it important to recycle batteries and panels?
5. Choose a side for the debate ("Solar is 100% friendly") and write one paragraph (3-4 sentences) arguing your position.

WEEK 10: FINAL PROJECT AND DEMONSTRATION

Topic: Final Project and Demonstration

Breakdown (Subtopics):

- Group project: Designing a mini solar system for home lighting
- Component selection and installation steps
- Testing and presentation of group projects
- Peer review and instructor feedback

Performance Objectives: By the end of this lesson, students should be able to:

1. Work in a group to conduct a simple load audit.
2. Use the sizing formulas (from Week 6) to size a mini-system.
3. Select the correct components for their design (e.g., battery type, inverter type).
4. Create a wiring diagram for their system.
5. Present and defend their design to the class.

Lesson Content:

1. Project Brief: "The 3-Room House"

- **Goal:** In groups of 4-5, you will design a *complete* small solar system for a client's 3-room home.
- **The Load (The Client's Needs):**
 - 3 x 9W LED Bulbs (used for 6 hours/night)
 - 1 x 70W Ceiling Fan (used for 8 hours/night)
 - 1 x 80W Television (used for 5 hours/night)
 - 1 x 10W Phone Charger (used for 4 hours/night)
- **Constraints:**
 - System Voltage: 12V
 - Battery Type: Lead-Acid (50% DoD)
 - Days of Autonomy: 1 day

- Location: Nigeria (4.5 Peak Sun Hours)

2. Project Steps (To be done in groups):

- **Step 1: Load Audit (Calculation)**
 - Calculate the **Total Running Watts (W)**.
 - Calculate the **Total Daily Energy (Wh)**.
- **Step 2: System Sizing (Calculation)**
 - Calculate the required **Inverter Size (W)**.
 - Calculate the required **Battery Bank Size (Ah)**.
 - Calculate the required **Solar Panel Array (Wp)**.
- **Step 3: Component Selection (Making Choices)**
 - Based on your sizing, *choose* the actual components.
 - *Inverter*: What size? PSW or MSW? (Must be PSW!)
 - *Battery*: How many? (e.g., "We need 243Ah, so we will use *two* 200Ah batteries in parallel.")
 - *Panels*: How many? (e.g., "We need 421W, so we will use *two* 250W panels in parallel.")
 - *Controller*: PWM or MPPT? (MPPT is better!)
- **Step 4: Wiring Diagram**
 - Draw a neat, labeled diagram showing all your chosen components connected correctly (showing all series/parallel and AC/DC connections).
- **Step 5: Presentation**
 - Each group will present their design (their calculations and choices) to the class. Be prepared to *defend* your choices (e.g., "We chose a PSW inverter because the client has a TV," "We chose two batteries to meet the energy demand").

3. Peer Review and Feedback

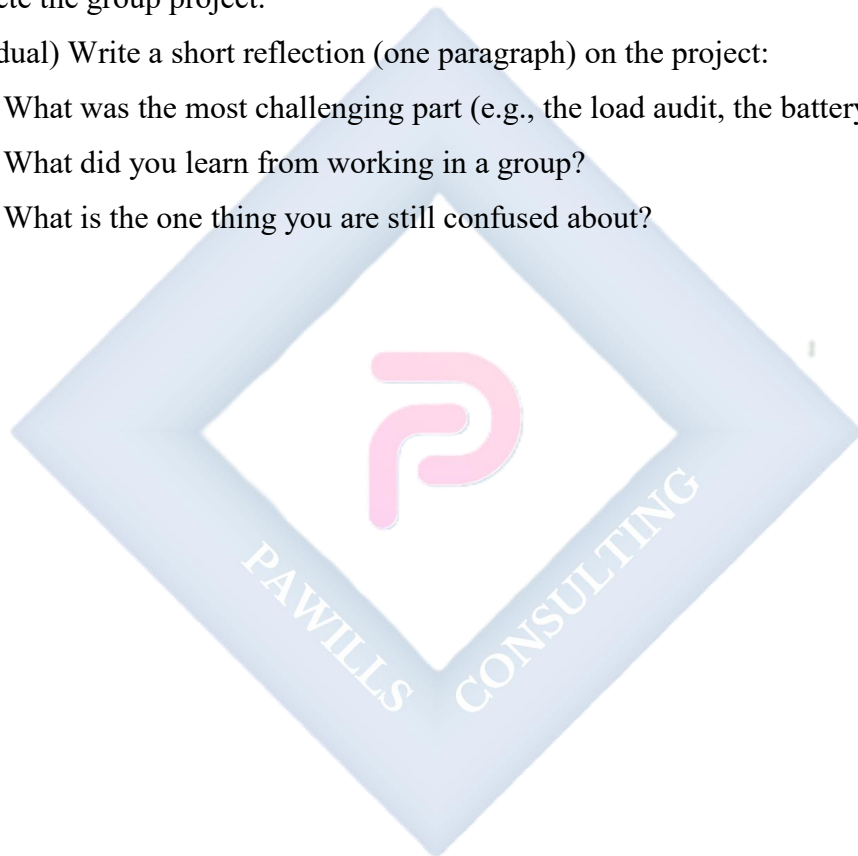
- (After each presentation, other students will ask questions and provide feedback, e.g., "Is your inverter large enough?" "What about your cable sizes?")

Evaluation: The evaluation for this week *is* the group project (presentation and calculations).

- Correctness of Load Audit (20%)
- Correctness of Sizing Calculations (30%)
- Correct Component Selection (20%)
- Clarity of Wiring Diagram (10%)
- Clarity of Presentation (20%)

Assignment:

1. Complete the group project.
2. (Individual) Write a short reflection (one paragraph) on the project:
 - What was the most challenging part (e.g., the load audit, the battery calculation)?
 - What did you learn from working in a group?
 - What is the one thing you are still confused about?



WEEK 11: REVISION

Topic: Revision

Breakdown (Subtopics):

- Comprehensive review of all topics covered

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from Term 2 and Term 3.
2. Mentally (or on paper) size a simple solar system from a given load.
3. Recall all major safety, installation, and maintenance procedures.
4. Discuss the career and environmental aspects of the solar industry.

Lesson Content:

1. Summary of Term 3 Topics (The "Advanced" Term):

- **Week 1 (Review):** Recapped Term 2 (Sizing, Batteries, Inverters, Safety).
- **Week 2 (Storage):** Energy storage solves the *intermittency* of renewables. We use **Chemical** (Batteries), but large-scale options include **Mechanical** (Pumped Hydro) and **Thermal** (Molten Salt). **Round-trip efficiency** (e.g., 95% for Lithium) measures energy loss.
- **Week 3 (Evaluation):** We test a system's performance by measuring **Voc** (Open Circuit Voltage) and **Isc** (Short Circuit Current) with a multimeter and comparing it to the datasheet.
- **Week 4 (Troubleshooting):** We find faults logically.
 - **Common Faults:** **Shading** (kills power), **Loose Connections** (starts fires), **Battery Failure** (dead), **Reversed Polarity** (destroys components).
 - **Tool:** Multimeter.
- **Week 6 (Entrepreneurship):** A solar entrepreneur (Installer, Seller, Maintainer) must have **Technical Skills** (sizing, wiring) and **Soft Skills** (Honesty, Communication, Punctuality).

- **Week 8 (Careers):** Many careers: **Technician** (hands-on), **Engineer** (design), **Scientist** (R&D), **Sales**. All require **STEM** skills.
- **Week 9 (Environment):**
 - **Pros:** Fixes climate change (no CO₂), no noise/air pollution.
 - **Cons:** Manufacturing pollution and **End-of-Life (EoL)** waste (toxic batteries/panels).
 - **Solution: Recycling.**
- **Week 10 (Project):** We put all our knowledge together (Audit -> Sizing -> Selection -> Diagram) to design a complete system.

2. Class Activity: Final Q&A "Hot Seat" (Teacher asks random students rapid-fire questions from all three terms to prepare for the final examination).

- "What are the 4 main components?"
- "Series vs. Parallel: which one adds voltage?"
- "What is DoD?"
- "What is the difference between a 500W Inverter and 500Wh of energy?"
- "What is the most dangerous safety hazard?"
- "What are the 3 main types of energy storage?"
- "What is the main negative impact of solar?"
- "What is the most important soft skill for an entrepreneur?"

Evaluation: This week's evaluation is the Third Term (Final) Examination, covering all three terms.

Assignment:

1. Study all notes from First, Second, and Third Term for the final examination.
2. Review all your calculations for Load Audits and System Sizing.
3. Draw a complete hybrid solar system diagram from memory.
4. Make a list of 10 key safety and maintenance rules.
5. Write one paragraph on why you think learning solar PV installation is (or is not) a valuable skill for your future.

